Task-Specific Technical Change and Comparative Advantage*

Lukas Althoff† Hugo Reichardt‡

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Abstract

Artificial intelligence changes work not only through augmentation and automation, but also through simplification: it can reduce the skills required to perform tasks well. We develop and estimate a dynamic task-based model in which workers accumulate multi-dimensional skills that shape their comparative advantage across tasks and, in turn, their occupational choices. We ground workers' evolving comparative advantage in the match between skills and skill requirements, and use the quantified model to predict how AI affects workers in general equilibrium. Our key finding is that AI substantially reduces wage inequality while raising average wages by 21 percent. The equalizing force is simplification, compressing wage differences by enabling workers across skill levels to compete for the same jobs. We document unfolding trends in labor markets, job postings, and college major choices consistent with our predicted effects of AI.

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†Stanford University. lalthoff@stanford.edu

‡CREi and Barcelona School of Economics. hreichardt@crei.cat

1 Introduction

Artificial intelligence is transforming how people work—reshaping the tasks they perform, the occupations they choose, and the skills they need. A quantitative framework is needed to understand how these task-level changes affect workers' careers and the overall labor market. Such a framework must capture how workers build the skills that shape their productivity across different tasks, how workers respond to technological change by switching occupations, and how wages adjust to reflect these shifts. This paper develops and estimates a dynamic task-based model of the labor market, allowing researchers to estimate the general equilibrium effects of any task-specific technical change—observed or counterfactual. We apply this methodology to study the labor market effects of artificial intelligence.

We model technical change as the augmentation, automation, and simplification of tasks. Augmentation and automation are standard features of task-based models, capturing (skill-neutral) increases in human productivity and the substitution of labor by capital (e.g., Acemoglu and Autor, 2011; Acemoglu and Restrepo, 2018, 2019). Beyond those standard forces, we introduce *simplification*, reflecting that technologies can reduce the skills required to complete a task—equivalently, augmenting low-skill workers more than high-skill workers. Recent experimental evidence shows that less-skilled workers experience larger AI-induced productivity gains, highlighting the empirical relevance of this channel (e.g., Dell'Acqua et al., 2023; Noy and Zhang, 2023; Brynjolfsson et al., 2025b; Cruces et al., 2026).

Understanding how task-specific technical change affects workers requires a structural framework. To see why, consider radiologists, an occupation where AI is reshaping tasks in multiple ways: AI can augment screening, automate detection, and simplify reporting (Hosny et al., 2018; Eloundou et al., 2024; Mousa, 2025). Given these changes, how do wages and demand for radiologists adjust? Do their existing skills become more or less valuable? Could they transition to related medical specialties, and if retraining is required, how quickly can they accumulate new skills? Without answering such questions—jointly and in general equilibrium—one cannot predict how workers will be impacted by AI.

Our model combines the key features necessary to understand the labor market effects of task-specific technical change. Specifically, workers have multi-dimensional skills and their productivity depends on the match between these skills and the task's skill requirements. Workers accumulate skills by performing tasks where requirements exceed their current skills. Each period, workers choose an occupation (a bundle of tasks) considering the future benefits of learning on the job.

To estimate the model, we must identify (i) how skills determine workers' productivity across tasks and (ii) how workers accumulate skills over their careers. The assumption that workers' time is optimally allocated across tasks yields a closed-form mapping from task-level productivity to occupation-level productivity and observed wages. We recover task-level productivity from this mapping combined with detailed data on workers' skills, occupations, and wages as well as tasks and the skills they require. We identify skill accumulation from workers' occupational histories and the evolution of their wages.

We estimate the model parameters that govern workers' productivity, learning, and occupational choices by maximum likelihood using data from the National Longitudinal Survey of Youth 1979 (NLSY79). In our quantitative application, workers' skills are five-dimensional (manual, social, math, technical, verbal) and workers differ in their "ability to learn" (the rate at which they accumulate skills). We develop a procedure that is computationally feasible despite the dynamic and high-dimensional nature of the model. First, we recover equilibrium prices from data directly, avoiding the need to solve the model within the estimation routine. Second, we exploit that many parameters can either be estimated by linear regression or be obtained with a fast iterative routine conditional on the remaining parameters, substantially reducing the dimension of the parameter space that must be searched non-linearly.

Our quantified model offers a laboratory to study how any counterfactual task-specific technical change affects the labor market. The model predicts how technologies change workers' comparative advantage, how workers reallocate and retrain, and how prices adjust in general equilibrium. We provide computationally efficient algorithms to solve for the new steady state and the transition path. We do so by simulating workers' choices and skill accumulation, updating prices iteratively to equate demand and supply.

We combine our model with information on AI's technological capabilities to predict how individual workers and labor markets as a whole will be affected. We measure AI's capabilities to augment, automate, and simplify each task by using large language models to scale established survey designs (following Eloundou et al., 2024; Aghion and Bunel, 2024; Acemoglu, 2025). While inherently uncertain, we show that the LLM-generated estimates of augmentation and automation align with human expert assessments and experimental estimates of AI-effects across various tasks and occupations. We also show that direct measures of occupational skill requirements derived from job postings data have decreased in the occupations for which the LLM predicts simplification by AI.[1] Our baseline analysis focuses on the impact of generative AI, but we also consider AI technologies with physical capabilities.

We find that AI has large positive effects, reducing wage inequality and increasing average wages by 21 percent. The key force behind the reduction in inequality is simplification. It lowers skill-based barriers by increasing the relative productivity of lower-skill workers in tasks and occupations that were previously the territory of higher-skilled workers. Automation and augmentation, on the other hand, have quantitatively small distributional implications, but drive most of the average wage increase. Simplification, in turn, does not affect the average wage by much because two opposing forces roughly cancel out: it increases productivity for given skills but it reduces skill accumulation. A large literature has documented that technical change over recent decades has increased the demand for skilled workers and widened wage inequality (Katz and Murphy, 1992; Katz and Autor, 1999; Goldin and Katz, 2008). Our finding marks a departure from this pattern, driven by simplification reversing the direction of skill bias.

Second, we find that AI significantly alters the occupational landscape. For example, administrative occupations (e.g., financial clerks) see a large decline in employment, while science occupations (e.g., life scientists) expand. Wages rise on average but some occupations—such as architects, engineers, and executives—see absolute wage declines. In many cases, the occupations that experience the largest employment gains are also those for which relative wages decrease the most. This negative relationship between wage and employment effects arises from simplification, which makes jobs easier to perform, enlarges the pool of qualified workers, and suppresses average wages through selection and increased competition (Autor and Thompson, 2025). In contrast, augmentation and automation barely affect relative wages as they leave workers' comparative advantage largely unchanged.

Third, while workers face a cost of adjustment to the new technology, we

[1] Also see Asirvatham et al. (2026) who find that using LLMs as a measurement tool "is accurate across domains and generally indistinguishable from human evaluators."

find that AI generates sizable welfare gains. Despite having to cope with the unexpected transition to the AI era, current workers experience welfare gains of 10 to 40 percent. These gains are largest for the lowest-skill workers who benefit disproportionately from simplification. Older workers face the greatest heterogeneity—some have skills and occupations that happen to match the needs of the AI era, while others do not. The skill that retains most value is math, while the returns to verbal skills strongly decline.

Lastly, early labor market evidence provides support for our model's predictions. Using an event study design, we find that the release of increasingly capable AI models after 2022 has gone hand-in-hand with the expansion of occupations predicted to benefit from AI, with 13 percent of our predicted effects materializing by early 2026. A second event study shows that college major choices have begun to respond consistently with our predicted changes in the labor market returns to skills built in those majors.

Related literature. To understand the labor market effects of task-based technical change, we integrate three literatures into one empirically tractable framework: task-based production, multi-dimensional skills, and dynamic occupational choice. First, we provide methods to estimate workers' comparative advantage across tasks—key for workers' ability to reallocate in response to technical change (Autor et al., 2003; Acemoglu and Autor, 2011; Autor and Handel, 2013; Acemoglu and Restrepo, 2018, 2022; Hurst et al., 2024). The absence of such methods has made it difficult to quantify the effects of counterfactual (future) technical change (Restrepo, 2024; Woessmann, 2024). Second, while most of the literature on task-based production treats workers' skills as fixed, we incorporate skill accumulation—also key for workers' adaptation. Third, we integrate task-based production into a general equilibrium dynamic occupational choice model (Keane and Wolpin, 1997; Heckman et al., 1998; Lee and Wolpin, 2006; Dix-Carneiro, 2014; Traiberman, 2019; Humlum, 2021; Smeets et al., 2025).

Second, we introduce simplification as a new channel of task-based technical change alongside automation and augmentation. Simplification reduces a task's skill requirements, increasing the relative productivity of lower skilled workers.[2] Autor and Thompson (2025) show that historically, the effect of automation on occupations' wages has depended on whether "expert" or "non-

[2]Simplification is thus a form of low-skill-biased technical change (Katz and Murphy, 1992).

expert" tasks are automated—a form of indirect simplification.[3] Downey (2021) and Danieli (2025) also consider mechanisms related to simplification.[4] We embed and quantify all three channels—augmentation, automation, and (direct) simplification—in a task-based general equilibrium framework and find that (direct) simplification drives AI's distributional effects.

Third, we advance our understanding of how workers' multi-dimensional skills shape their comparative advantage across tasks. Recent work has emphasized the multi-dimensionality of skills (Lindenlaub, 2017; Guvenen et al., 2020; Lise and Postel-Vinay, 2020; Baley et al., 2022; Ide and Talamas, 2026). Task-based models are inherently multi-dimensional because workers are characterized by their productivity across many tasks. Quantifying those models requires estimating each worker's productivity, including in tasks they do not currently perform. We overcome this identification problem by providing a microfoundation of task-level productivity as a function of workers' measured skills (e.g., manual, social, math) and tasks' skill requirements. We also provide a new database of such task-level skill requirements, expanding beyond the occupational aggregates available in O*NET (e.g., Lise and Postel-Vinay, 2020; Baley et al., 2022).

Finally, by applying our general task-based model to AI, we contribute to the effort of understanding the unfolding labor market impacts of this new technology. Freund and Mann (2025) introduce a framework to study how automation affects wages through changes in the importance of tasks within occupations. Our approach differs in three key ways: we model simplification as a distinct channel through which AI reshapes comparative advantage, we allow workers' skills to evolve over their careers (making the model dynamic), and we estimate task-level comparative advantage directly from workers' measured skills and tasks' skill requirements.[5] Hosseini and Lichtinger (2026) show that even a simple model—static and with one-dimensional skill—can already capture part of the simplification mechanism. Hampole et al. (2025) provide a structural framework to quantify AI's effect on occupational demand through observed adoption across firms. In contrast, our paper predicts how AI affects individual workers—who may change occupations.

[3] By indirect simplification we mean the occupation-level decrease in skills ("expertise") required by automating tasks with relatively high skill requirements.

[4] Simplification is also similar in spirit to "non-autonomous AI," which allows lower-skill workers to solve hard problems with AI, raising their relative wages (Ide and Talamàs, 2025).

[5] In contrast, Freund and Mann (2025) infer the distribution of task-level productivity indirectly from data on wages and occupational choices.

2 Model

We develop a model that describes how workers choose occupations, perform tasks, and accumulate skills over their careers. Overlapping generations of workers live for A periods. Productivity and wages depend on the match between workers' skills and the skill requirements of the tasks relevant to the occupation of choice. Workers accumulate skills on the job, so that their occupational choice depends on the current wage as well as the learning benefits that the job offers. Prices are determined in general equilibrium. Technical change can take the form of augmentation, automation, and simplification of tasks.

2.1 The Firm's and Worker's Problems

Occupations and tasks. Each occupation produces a distinct good by combining a unique set of tasks. These tasks are combined with a constant elasticity of substitution ρ, so that the production function of occupation j is

$$Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} y_\tau^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \tag{1}$$

where $\mathcal{T}_j$ is the discrete set of relevant tasks, y_τ is the output of task τ, and $\theta_{j,\tau}$ is task τ's importance weight that satisfies $\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} = 1$.

Task-level productivity and skills. The production function for task τ in occupation j depends on whether the task is automatable, i.e., $\tau \in \mathcal{A}_j$, or not, i.e., $\tau \in \mathcal{N}_j$:

$$y_\tau(\boldsymbol{h}, \ell_\tau, k_\tau) = \begin{cases} \ell_\tau \gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau) & \text{if } \tau \in \mathcal{N}_j \\ \ell_\tau \gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau) + \phi_\tau k_\tau & \text{if } \tau \in \mathcal{A}_j \end{cases} \tag{2}$$

where ℓ_τ represents the share of the worker's time allocated to task τ, γ_τ and ϕ_τ capture the task-specific productivity of humans and capital respectively, $\boldsymbol{h} = (h_s)'_{s \in S}$ is a vector capturing workers' multi-dimensional skills, $\boldsymbol{r}_\tau \equiv (r_{\tau,s})'_{s \in S}$ is the skill requirement of task τ, and k_τ is capital devoted to task τ. If the task is automatable, capital and labor are perfect substitutes.

The function $f(\cdot)$ determines how workers with different skills $\boldsymbol{h}$ are differentially productive in tasks depending on their skill requirements, $\boldsymbol{r}_\tau$. Intuitively, this function captures how workers' productivity depends on the "match" between their skills and the skills required to complete the task (Lise and Postel-Vinay, 2020; Baley et al., 2022). For our quantification, we will assume a functional form for $f(\cdot)$ (in section 2.4, equation 14) and show how its parameters can be identified and estimated.

Technical change. We consider three different ways in which technical change affects the task-level production function:

Augmentation	Enhancing human productivity, increasing γ_τ;
Automation	Substituting labor with capital, expanding $\mathcal{A}_j$;
Simplification	Simplifying tasks for humans, reducing $\boldsymbol{r}_\tau$.

The first two are standard in the task-based literature (e.g., Acemoglu and Autor, 2011). We introduce simplification to allow for heterogeneity in technologies' impact on productivity depending on workers' skills—essentially skill-biased augmentation at the task-level (Katz and Murphy, 1992).[6] Experimental evidence has shown that AI's productivity effects tend to be stronger for lower-skill workers, suggesting that simplification is an important force in practice (e.g., Dell'Acqua et al., 2023; Noy and Zhang, 2023; Brynjolfsson et al., 2025b; Cruces et al., 2026).

The firm's problem. Each good j is produced by a representative firm that takes the equilibrium wage $\{w_j(\boldsymbol{h})\}_{\boldsymbol{h}}$ and the costs of capital R as given. The firm chooses how many workers of each skill set $\boldsymbol{h}$ to hire, how to allocate their time across tasks, and how much capital to use for each task. Formally, the firm

[6] While we call this type of technical change "simplification," the methodology equally allows for increases in skill requirements (or "complication").

solves the following profit maximization problem:

$$\max_{\{n_j(\boldsymbol{h}),\ell_{j,\tau}(\boldsymbol{h}),k_{j,\tau}(\boldsymbol{h})\}} \int n_j(\boldsymbol{h}) \left(p_j Y_j(\boldsymbol{h}) - w_j(\boldsymbol{h}) - R \sum_{\tau \in \mathcal{A}_j} k_{j,\tau}(\boldsymbol{h}) \right) d\boldsymbol{h}$$

$$\text{s.t. } Y_j(\boldsymbol{h}) = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \tilde{y}_\tau(\boldsymbol{h})^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \tag{3}$$

$$\sum_{\tau \in \mathcal{T}_j} \ell_{j,\tau}(\boldsymbol{h}) = 1 \, \forall \boldsymbol{h}$$

$$\tilde{y}_\tau(\boldsymbol{h}) \equiv y_\tau(\boldsymbol{h}, \ell_{j,\tau}(\boldsymbol{h}), k_{j,\tau}(\boldsymbol{h})) \text{ given by (2)},$$

where $n_j(\boldsymbol{h})$ is the amount of labor employed with skill $\boldsymbol{h}$, $\ell_{j,\tau}(\boldsymbol{h})$ is the share of time allocated to task τ, and $k_{j,\tau}(\boldsymbol{h})$ denotes the capital per worker allocated to workers with skill $\boldsymbol{h}$ working on task τ. Because the marginal product of capital depends on a worker's skills, the firm generally does not allocate an equal amount of capital to each worker.

Since the labor market is perfectly competitive the worker's wage must equal their marginal product. That is,

$$w_j(\boldsymbol{h}) = p_j Y_j(\boldsymbol{h}) - R \sum_{\tau \in \mathcal{A}_j} k_{j,\tau}(\boldsymbol{h}). \tag{4}$$

A worker's wage thus depends on the allocation of their time and capital to the various tasks in the occupation. If the worker only cares about the allocation insofar as it affects the wage, the worker and firm would always agree that the allocation should maximize the worker's value added. However, when the allocation also affects future payoffs (such as through skill accumulation as will be the case in this model) the worker would in principle be willing to accept a lower wage in exchange for an allocation that yields higher future payoffs (through increased learning). To rule out such an exchange, we make the following assumption:

Assumption 1 (Control and non-contractibility of task assignment)**.** The firm controls the workers' time allocation across tasks which is not contractible.

Assumption 1 implies that the firm can freely determine how workers allocate their time across tasks. Because time allocation is non-contractible, any promised task assignment cannot be enforced by workers. As a result, the firm retains full flexibility to adjust workers' task allocations as it sees fit. This assumption implies that the firm chooses the allocation that maximizes profits (output net of capital costs) for any given wage $w_j(\boldsymbol{h})$. In equilibrium, wages must thus equal the worker's marginal product given this allocation.

A second assumption is that automatable tasks are cheaper to perform with capital than with labor, ensuring the worker's time is only allocated to non-automatable tasks:

Assumption 2 (Full automation of automatable tasks)**.** The unit cost of producing a task with capital is lower than the cost of producing it with labor for all occupations j, skills $\boldsymbol{h}$, and tasks $\tau \in \mathcal{A}_j$,

$$\frac{w_j(\boldsymbol{h})}{\gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau)} > \frac{R}{\phi_\tau}.$$

Optimal time allocation. Assumption 1 and 2 together imply that a worker's time is allocated to non-automatable tasks to maximize the production of these tasks. That is, the firm solves the following time allocation problem:

$$\{\ell_{j,\tau}(\boldsymbol{h})\}_{\tau \in \mathcal{N}_j} = \underset{\{\ell_\tau\}_{\tau \in \mathcal{N}_j}}{\arg\max} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \left(\ell_\tau \gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau)\right)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad \text{s.t.} \sum_{\tau \in \mathcal{N}_j} \ell_\tau = 1.$$

The solution to this problem for a given task $\tau \in \mathcal{N}_j$ is

$$\ell_{j,\tau}(\boldsymbol{h}) = \frac{\theta_{j,\tau} \gamma_\tau^{\rho-1} f(\boldsymbol{h}, \boldsymbol{r}_\tau)^{\rho-1}}{\sum_{\kappa \in \mathcal{N}_j} \theta_{j,\kappa} \gamma_\kappa^{\rho-1} f(\boldsymbol{h}, \boldsymbol{r}_\kappa)^{\rho-1}}, \tag{5}$$

which shows that more time is spent on tasks with greater weight $\theta_{j,\tau}$. If tasks are substitutes ($\rho > 1$), a worker's time is allocated to the tasks they are most productive in, i.e., tasks for which $\gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau)$ is greater. If instead tasks are complements ($\rho < 1$), workers spend more time on less productive tasks.

Optimal capital allocation. In choosing how much capital to allocate to each worker-task pair, the firm balances the marginal benefit of increased output against the cost of capital. The first order condition implies that for all tasks $\tau \in \mathcal{A}_j$

$$k_{j,\tau}(\boldsymbol{h}) = \theta_{j,\tau} \phi_\tau^{\rho-1} Y_j(\boldsymbol{h}) \left(\frac{p_j}{R}\right)^\rho, \tag{6}$$

where $Y_j(\boldsymbol{h})$ is the profit-maximizing level of output when a worker with skill vector $\boldsymbol{h}$ works in occupation j. Clearly, the lower the cost of capital relative to the price of the output, the more capital the firm uses. Also, firms allocate more task-automating capital to workers that are more productive in the non-automated tasks, i.e., workers for which $Y_j(\boldsymbol{h})$ is larger.

Occupational productivity. Given the optimal allocation of workers' time and capital to tasks, the output per worker of type $\boldsymbol{h}$ in occupation j is

$$\begin{aligned} Y_j(\boldsymbol{h}) &\equiv \left(\sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \left(\phi_\tau k_{j,\tau}(\boldsymbol{h})\right)^{\frac{\rho-1}{\rho}} + \sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \left(\ell_{j,\tau}(\boldsymbol{h})\gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau)\right)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \\ &= \Gamma_j^{\frac{\rho}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} f(\boldsymbol{h}, \boldsymbol{r}_\tau)^{\rho-1} \right)^{\frac{1}{\rho-1}} \end{aligned} \tag{7}$$

where

$$\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$$

is the labor share. Equation (7) thus shows that a worker's output in occupation j is a weighted average of their productivity across all non-automated tasks $\tau \in \mathcal{N}_j$.

Wages. Combining equations (4), (6), and (7) yields the wage when a worker of skill $\boldsymbol{h}$ chooses occupation j:

$$w_j(\boldsymbol{h}) = p_j Y_j(\boldsymbol{h}) \Gamma_j = p_j \Gamma_j^{\frac{1}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} f(\boldsymbol{h}, \boldsymbol{r}_\tau)^{\rho-1} \right)^{\frac{1}{\rho-1}}. \tag{8}$$

Equation (8) shows that if none of the tasks are automatable, i.e. $\mathcal{A}_j = \varnothing$ and $\Gamma_j = 1$, a worker's income equals total revenue $w_j(\boldsymbol{h}) = p_j Y_j(\boldsymbol{h})$.

Skill accumulation. Before entering the labor market at age $a = 1$, each worker draws an initial skill vector $\boldsymbol{h}_1$ after which they accumulate further skills on the job. We assume that a worker's human capital accumulation depends on their current skills, their "ability to learn" ψ, and the skill requirements of the tasks in their job j: $\boldsymbol{h}' = g_j(\boldsymbol{h}, \psi)$.

Occupational choice. Every period, each worker chooses from a discrete set of occupations to maximize utility. Workers consume their income each period and live for A periods. The expected lifetime utility of a worker aged a, with learning ability ψ, and previous occupation k is represented by the value function

$$V_a(\boldsymbol{h},\psi,k) = \mathbb{E}\left[\max_j \log w_j(\boldsymbol{h}) + \log \varepsilon_j + \mu_j - \kappa(k,j) + \beta V_{a+1}\left(g_j\left(\boldsymbol{h},\psi\right),\psi,j\right)\right] \tag{9}$$

where $\mathbb{E}[\cdot]$ is the expectation over the occupation-specific productivity shocks ε_j. These productivity shocks are relevant to wages: a worker's observed wage equals the deterministic part ($w_j(\boldsymbol{h})$) times the stochastic part (ε_j).

Each occupation j has an amenity value μ_j. $\kappa(k,j)$ is a cost of switching from occupation k to j. In our quantitative application, we set this to $\kappa(k,j) = \kappa\mathbb{1}[j \neq k]$ for some constant κ. $g_j\left(\boldsymbol{h},\cdot\right)$ is next period's human capital when choosing occupation j. Because the worker's life is finite, the value after terminal age A is zero, $V_{A+1}(\cdot) = 0$.

We assume that the productivity shocks $\log \varepsilon_j$ follow a type I generalized extreme value (Gumbel) distribution with mean 0 and scale parameter ζ.[7] This assumption implies that the conditional probability of choosing occupation j has the closed-form solution

$$\mathbb{P}_a(j \mid \boldsymbol{h},\psi,k) = \frac{\exp\left(\frac{1}{\zeta}\left(\log w_j(\boldsymbol{h}) + \mu_j - \kappa(k,j) + \beta V_{a+1}\left(g_j\left(\boldsymbol{h},\psi\right),\psi,j\right)\right)\right)}{\sum_{l=1}^{J}\exp\left(\frac{1}{\zeta}\left(\log w_l(\boldsymbol{h}) + \mu_l - \kappa(k,l) + \beta V_{a+1}\left(g_l\left(\boldsymbol{h},\psi\right),\psi,l\right)\right)\right)} \tag{10}$$

so that the value function in (9) can be simplified to

$$V_a(\boldsymbol{h},\psi,k) = \zeta \log \sum_{j=1}^{J} \exp\left(\frac{1}{\zeta}\left(\log w_j(\boldsymbol{h}) + \mu_j - \kappa(k,j) + \beta V_{a+1}\left(g_j\left(\boldsymbol{h},\psi\right),\psi,j\right)\right)\right) \tag{11}$$

Since $V_{A+1}(\cdot) = 0$, equation (11) solves the value function, and thus the occupational choice problem, by backward iteration from age A to 1 for a given sequence of prices. Note that the notation above omits any dependence on time. In principle, prices vary over time, so that the wage schedule $w_{j,t}(\boldsymbol{h})$, and thus the value functions, are time-dependent.

[7] The CDF is $\Pr(\log \varepsilon < x) = \exp\left(-\exp\left(-\frac{x+\zeta\overline{\gamma}}{\zeta}\right)\right)$ where $\overline{\gamma} \approx 0.577$ is Euler's constant. This implies that ε_j follows a Fréchet distribution.

2.2 Equilibrium

The price of each occupational good p_j is determined in equilibrium through demand and supply. The supply is characterized by the solution to the worker's problem. The workers, in turn, consume and generate demand for the occupational goods. We assume that demand is characterized by a homothetic and invertible demand function $D\left(\{p_j\}_{j=1}^J\right)$ that maps prices p_j into relative demand for each occupational good. In our application, we use (standard or nested) CES demand. Having specified demand, we can now define the competitive equilibrium.

Definition (Competitive equilibrium)**.** Given an initial joint distribution of age, skills, ability, and occupations, $G_{a,t}(\boldsymbol{h},\psi,k)$, a distribution of human capital at birth $G_{1,t}(\boldsymbol{h},\psi)$, and the supply of capital $\{\mathcal{K}_t\}_{t=1}^{\infty}$, a *competitive equilibrium* is defined as a sequence of prices $\left\{p_{1,t},\cdots,p_{J,t},R_t\right\}_{t=1}^{\infty}$ such that

- Workers' occupational choices maximize the present value of lifetime utility given the sequence of prices. That is, their occupational choice probabilities are as in equation (10);

- The distribution over states follows from occupational choices:

$$G_{a+1,t+1}\left(\boldsymbol{h}',\psi,j\right)=\sum_{k=1}^{J}\int_{g_j(\boldsymbol{h}',\psi)\leq\boldsymbol{h}}\mathbb{P}_{a,t}\left(j\mid\boldsymbol{h},\psi,k\right)dG_{a,t}\left(\boldsymbol{h},\psi,k\right); \qquad (12)$$

- Demand for goods equals supply: $D(\{p_{j,t}\}_{j=1}^J)\propto\mathcal{Y}_{j,t}$, where

$$\mathcal{Y}_{j,t}\equiv\sum_{a=1}^{A}\sum_{k=1}^{J}\int Y_j(\boldsymbol{h})\,\mathbb{P}_{a,t}(j\mid\boldsymbol{h},\psi,k)\,\mathbb{E}[\varepsilon_j\mid j,\boldsymbol{h},\psi,k]\,dG_{a,t}(\boldsymbol{h},\psi,k);^{8} \qquad (13)$$

- Demand for capital equals supply:

$$\mathcal{K}_t=\sum_{a=1}^{A}\sum_{j=1}^{J}\sum_{\tau\in\mathcal{A}_j}\int k_{j,\tau}(\boldsymbol{h})\mathbb{P}_{a,t}\left(j\mid\boldsymbol{h},\psi,k\right)dG_{a,t}\left(\boldsymbol{h},\psi,k\right).$$

[8] $\mathbb{E}[\varepsilon_j\mid j,\boldsymbol{h},\psi,k]$ is the expectation of the productivity shock conditional on choosing occupation j when your states were $\boldsymbol{h},\psi,k$. The Gumbel distribution of $\log\varepsilon_j$ implies that this expectation has a closed-form solution: $\mathbb{E}[\varepsilon_j\mid j,\boldsymbol{h},\psi,k]=\exp(-\zeta\gamma)\Gamma(1-\zeta)\mathbb{P}_{a,t}(j\mid\boldsymbol{h},\psi,k)^{-\zeta}$.

2.3 Solution Methods

We provide algorithms to solve for a stationary competitive equilibrium as well as the transition path after an unexpected one-off technological shock.

Stationary equilibrium. To solve for a stationary equilibrium, we use the following algorithm:

1. Guess an initial vector of relative prices $\left(p_1^{(1)}, \cdots, p_J^{(1)}\right)$.

2. For iteration r, given the prices $\left(p_1^{(r)}, \cdots, p_J^{(r)}\right)$, solve the worker's problem and compute the implied output of each good $\left(\mathcal{Y}_1^{(r)}, \cdots, \mathcal{Y}_J^{(r)}\right)$. Then, update prices to clear the market given supply: $\boldsymbol{p}^{(r+1)} = D^{-1}\left(\{\mathcal{Y}_j^{(r)}\}_{j=1}^{J}\right)$.[9]

3. Repeat step 2 until $\left\|\boldsymbol{p}^{(r+1)} - \boldsymbol{p}^{(r)}\right\| < \epsilon$ for a threshold $\epsilon > 0$.

Transition path. Starting from the initial stationary equilibrium, we solve for the transition path of prices $\{p_{1,t}, \cdots, p_{J,t}\}_{t=1}^{\infty}$ from the moment the shock is realized. We numerically approximate this infinite sequence by solving for $\{p_{1,t}, \cdots, p_{J,t}\}_{t=1}^{T}$ for a large enough T such that prices are constant after period T. The solution algorithm is based on (Boppart et al., 2018):

1. Compute the stationary equilibrium before ($t = 0$) and after the change ($t = T$).

2. Guess a path for the sequence of prices.[10]

3. For iteration r, given the sequence of prices $\left\{p_{1,t}^{(r)}, \cdots, p_{J,t}^{(r)}\right\}_{t=1}^{T}$, solve for the value function at $t = T, T-1, \ldots, 1$. Then, compute the implied output of each good at each time and the corresponding prices $\boldsymbol{p}_t^{(r+1)} = D^{-1}\left(\{\mathcal{Y}_{j,t}^{(r)}\}_{j=1}^{J}\right)$ for $t = 1, \ldots, T$.

4. Repeat step 3 until $\left\|\boldsymbol{p}_t^{(r+1)} - \boldsymbol{p}_t^{(r)}\right\| < \epsilon \; \forall t = 1, \ldots, T$ for a threshold $\epsilon > 0$.

[9] In case of simple CES demand for occupations as in our baseline, there is a simple closed-form solution for the updated price. We also consider a nested CES structure, in which case the price update can be computed using a Newton tâtonnement algorithm.

[10] A reasonable guess is the path where prices adjust immediately to the new stationary equilibrium.

Computing implied output given the sequence of prices is the main computational challenge in these algorithms. It consists of three main steps for which we provide more detail below.

First, we solve for the value function. This step is conceptually straightforward. However, when skills are multi-dimensional and there are many occupations j, the state space $(\boldsymbol{h}, \psi, j)$ is large and value function iteration costly. We exploit a convenient feature of the value function in equation (11) to provide relief. Since the occupational switching cost is $\kappa(j,k) = \kappa 1[j \neq k]$, the value function can be written as

$$V_a(\boldsymbol{h}, \psi, k) = \zeta \log \left[e^{-\frac{\kappa}{\zeta}} \sum_{j=1}^{J} \tilde{V}_a^j(\boldsymbol{h}, \psi) + \left(1 - e^{-\frac{\kappa}{\zeta}}\right) \tilde{V}_a^k(\boldsymbol{h}, \psi) \right]$$

where $\tilde{V}_a^j(\boldsymbol{h}, \psi) \equiv \exp\left(\frac{1}{\zeta}\left(\log w_j(\boldsymbol{h}) + \mu_j + \beta V_{a+1}\left(g_j(\boldsymbol{h}, \psi), \psi, j\right)\right)\right)$. This solution implies that it is sufficient to solve for $\tilde{V}_a^j(\boldsymbol{h}, \psi)$. Using this property, we effectively shrink the state space in the value function iteration step by a factor equal to the number occupations J (in our exercise, $J = 93$). Conditional choice probabilities can then be recovered as

$$\mathbb{P}_a(j \mid \boldsymbol{h}, \psi, k) = \frac{\tilde{V}_a^j(\boldsymbol{h}, \psi)\, e^{-\frac{\kappa}{\zeta} 1[j \neq k]}}{e^{-\frac{\kappa}{\zeta}} \sum_{j=1}^{J} \tilde{V}_a^j(\boldsymbol{h}, \psi) + \left(1 - e^{-\frac{\kappa}{\zeta}}\right) \tilde{V}_a^k(\boldsymbol{h}, \psi)}.$$

Second, after computing the value function and conditional choice probabilities, we compute the joint distribution of age a, skills $\boldsymbol{h}$, learning ability ψ, and occupations k using the law of motion in equation (12). We do so by simulation. That is, we first draw from the initial distribution of skills and learning ability at age $a = 1$. Their states imply a conditional probability to choose each occupation. We then draw occupational choices randomly based on these probabilities to obtain the distribution at age $a = 2$. We iterate this process forward until $a = A$.[11]

Third, to obtain an update for the relative prices, we compute total implied production of each good given the previous price iteration. The previous steps yield a sample of workers with a given occupation and set of skills. From there, we approximate the integral in equation (13) for each occupation j. That is,

[11]To save computational costs at early price iterations, we begin with a small number of simulations, and increase sample sizes as the difference between subsequent price iterations decrease.

we evaluate the term $Y_j(\boldsymbol{h})\,\mathbb{P}_{a,t}(j \mid \boldsymbol{h}, \psi, k)\,\mathbb{E}[\varepsilon_j \mid j, \boldsymbol{h}, \psi, k]$ for each worker-age and for each $j = 1, \ldots, J$. We do not condition on the occupational draw in the computation of production so that sampling noise only affects workers' states, not production conditional on those states.

2.4 Parametrization

In our quantitative application, we make assumptions on the functions that govern task-level productivity and skill accumulation.

Production. We specify the task-level production function as

$$f(\boldsymbol{h}, \boldsymbol{r}_\tau) = \prod_{s \in S} h_s^{\omega_s} \exp\left(-\eta \min\left\{h_s - r_{\tau,s}, 0\right\}^2\right). \tag{14}$$

This production function is similar to that proposed by (Lise and Postel-Vinay, 2020). The first term in equation (14) reflects a force that makes workers with higher skills more productive in *any* task, independent of its skill requirements. The second (exponential) term captures the degree to which the worker's productivity is diminished when performing tasks for which they are "underqualified." Figure A.1a shows the functional form graphically.

This functional form assumption implies that the wage function in (4) equals

$$w_j(\boldsymbol{h}) = p_j \Gamma_j^{\frac{1}{1-\rho}} \prod_{s \in S} h_s^{\omega_s} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \exp\left(-\eta \sum_{s \in S} \min\left\{h_s - r_{\tau,s}, 0\right\}^2\right)^{\rho-1} \right)^{\frac{1}{\rho-1}}. \tag{15}$$

Skill accumulation. We assume that skill accumulates according to the following functional form:

$$g_{j,s}(\boldsymbol{h}, \psi) = (1-\delta) h_s + \sum_{\tau \in \mathcal{N}_j} \ell_{j,\tau}(\boldsymbol{h}) \max\left\{r_{\tau,s} - h_s, 0\right\} e^{-\lambda(\psi) \max\{r_{\tau,s} - h_s, 0\}}, \tag{16}$$

where $\ell_{j,\tau}(\boldsymbol{h})$ indicates workers' time spent on task τ, defined in equation (5). Equation (16) has several intuitive implications for skill accumulation. First, workers' learning is most affected by the tasks they spent most time on, i.e., for which $\ell_{j,\tau}(\boldsymbol{h})$ is greatest.[12] Second, workers learn by performing tasks that are "hard" for them, i.e., tasks that have skill requirements above their current skill levels. However, workers learn most from tasks that are not "too hard." As tasks become harder relative to the workers' skill, the rate at which skills catch up decreases; $\lambda(\psi) \geq 0$ governs the rate of this slowdown. Figure A.1b illustrates how learning varies with the distance between the worker's skills and the task's requirements.[13] Similar to Heckman et al. (1998), we allow for workers to differ in their ability to learn ψ. Lastly, some skill depreciation occurs independently of which tasks are performed, governed by δ.

3 Data

We use three main data sources to estimate the model's parameters. First, we rely on O*NET to measure each occupation's tasks and skill requirements. Second, we provide and validate a new database that extends O*NET's occupation-level data on skill requirements to the task level using large language models. Third, we use panel data on wages, occupational choices, and multi-dimensional skills from the NLSY79.

For the application to AI, we also require data on AI's capability to augment, automate, and simplify each task. We follow the literature in using large language models to estimate these capabilities (e.g., Eloundou et al., 2024; Acemoglu, 2025). We compare our estimates to human expert ratings and experimental evidence, and provide additional validation through job postings data.

3.1 Estimation Data

3.1.1 Occupations and Tasks (O*NET)

O*NET is the leading database on occupations, tasks, and skills in the US economy (e.g., Autor et al., 2003; Acemoglu and Autor, 2011; Lise and Postel-Vinay, 2020). O*NET contains detailed descriptions of 19,530 tasks linked to 974 occupations. We rely on these data to define both the occupations j and tasks $\mathcal{T}_j$

[12] The workers' time allocation across tasks thus affects skill accumulation. Assumption 1 implies that firms do not take this into account.

[13] Since $f(x) = x\exp(-\lambda x)$ is strictly increasing for $x < \frac{1}{\lambda}$ and strictly decreasing after, learning from task τ is maximized when $r_{\tau,s} - h_s = 1/\lambda$, yielding a learning gain of $1/(e\lambda)$.

in our model.[14] We set the weights of each task τ in an occupation, $\theta_{j,\tau}$, to the importance measure of that task as reported in O*NET.

We also use O*NET's definition of worker skills across 35 dimensions (e.g., "reading comprehension" or "social perceptiveness"). O*NET rates skills on a scale from 1 to 7 and provides anchors for each level (e.g., level 2 in reading comprehension means being able to "read step-by-step instructions for completing a form" and 4 to "understand an email from management describing new personnel policies").[15]

We reduce O*NET's dimensions into five skill categories: manual, math, social, technical, and verbal.[16] Table B.1 shows this mapping.

3.1.2 Task-level Skill Requirements

Workers' comparative advantage in a task is governed by the match between their skills and the task's skill requirements, $\boldsymbol{r}_\tau$. While O*NET provides data on occupation-level skill requirements, it lacks task-specific data. To address this gap, we use OpenAI's GPT-4o to estimate such task-level skill requirements. To ensure consistency with O*NET and a valid survey design, we replicate O*NET's occupation-level questionnaire at the task level, by using their questionnaire format, skill dimensions, and skill anchors. This process covered 19,530 task descriptions across 35 skills using 683,550 queries. See Appendix D.1 for further details on our prompt design.

We validate our data by comparing aggregations of our newly generated task-level data with O*NET's occupation-level measures. For each occupation, we calculate importance-weighted average task-level skill requirements ($\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} r_{\tau,s}$) and compare these with corresponding O*NET values. The five aggregated skills have high agreement rates, with correlations ranging from 0.82 to 0.93 (see Figure A.2).[17] The high agreement with O*NET suggests that the LLM accurately identifies differences in skill requirements across occupations.

[14] Instead of the most granular definition of occupations, our analysis will rely on 93 aggregate (3-digit) occupation groups.

[15] The original O*NET questionnaire and skill level descriptions are available here.

[16] We equally weight each component in the aggregation of skills to our five final dimensions. Relative to Addison et al. (2020); Baley et al. (2022); DeLoach et al. (2022), we include manual as a separate skill because we believe its interaction with AI is of particular interest.

[17] Agreement rates are also high across most of the 35 original O*NET skill dimensions (see Figure A.3).

We also collect new human expert ratings of skill requirements for 200 skill-task combinations for the occupation *Economist*. Skill requirements are highly correlated between the LLM and the average of three human experts ($\rho = 0.76$; see Figure A.4a), with correlations between the LLM and individual experts ranging from 0.60 to 0.64. This correlation provides evidence that the LLM can also accurately distinguish skill requirements of different tasks within an occupation.

3.1.3 Skills, Occupational Choice, and Wages (NLSY79)

We use data from the NLSY79 to estimate the task-level production function and the skill accumulation function. The data contain information on wages, occupations, and multi-dimensional skill assessment scores.

We follow the literature in measuring skills in the NLSY79. As Addison et al. (2020); Baley et al. (2022), we measure skills with the Armed Services Vocational Aptitude Battery (ASVAB): (i) manual skills are measured as the average of standardized scores on *auto and shop information* and *mechanical comprehension*, (ii) math skills are based on *mathematics knowledge* and *arithmetic reasoning* scores; (iii) technical skills on *general science* and *electronics information*; (iv) verbal skills on *paragraph comprehension* and *word knowledge*.[18] For social skills, we use a composite measure of self-reported sociability as a young adult, sociability at age 6, the *Rotter Locus of Control Scale*, and the *Rosenberg Self-Esteem Scale* (see also Deming, 2017; Addison et al., 2020; Guvenen et al., 2020).

These data only provide an ordinal measure of skills. That is, we only observe $\tilde{\boldsymbol{h}} = F(\boldsymbol{h})$ where $F(\cdot)$ is the initial skill distribution. We do not directly observe the cardinal measure $\boldsymbol{h}$ that is on the same scale as the skill requirements. We recover the marginal distribution of skills by estimating it together with all other parameters (see section 4).

We follow the NLSY79 cohort's labor market history from age 25 to the survey in 2022. We retain information on all jobs held, including their start and end dates, the occupational code, the hourly wage, and the number of hours worked per week. Similar to Lise and Postel-Vinay (2020), we only consider workers for whom the maximum gap between observed jobs is no larger than 18 months. We collapse this data to a worker panel of yearly frequency.

[18]Relative to Addison et al. (2020) and Baley et al. (2022), we add manual skills as a separate dimension.

3.2 Data on AI's Capabilities

In the model presented in section 2, technical change can take three forms: augmentation, automation, and simplification. We estimate AI's capabilities across those three channels for each task. While there is inherent uncertainty surrounding these capabilities, we verify that our estimates align with experimental estimates, human expert opinions, and real-time evidence from job postings data. As our baseline, we consider the effects of generative AI (data summarized in Table B.3).[19] We also consider AI-powered robots in an alternative scenario.[20]

Augmentation. In measuring AI's potential to augment human productivity, we follow Eloundou et al. (2024) who asked human raters and OpenAI's GPT-4 whether they believed that LLMs can reduce the worker's time required to complete a task by at least half. We replicate their exercise with GPT-4o, except that we asked for a continuous estimate of the percentage of time saved (rather than a binary measure), and consider generative AI more broadly (rather than LLMs). On average, we estimate that generative AI saves 20.2 percent of worker time (see Table B.3). Appendix D.2.1 describes our prompt design. The prompt also describes how we extend it to AI-powered robots.

We validate our new data on task-level AI augmentation in two ways. First, we find that our estimates are strongly correlated with both the human rated and GPT-4 rated data from Eloundou et al. (2024)—see Figure A.5a. Second, we compiled experimental estimates of average productivity effects of generative AI in various tasks and occupations and compared them with our estimates (see Table B.4). Our estimates closely approximate those experimental estimates.

Average worker productivity in a task can be affected through both augmentation (making all workers equally more productive) and simplification (making less-skilled workers differentially more productive). To recover augmentation from aggregate productivity effects, we first compute simplification's average productivity effects across all workers.[21] We then compute augmentation's productivity effect as the difference between the aggregate and the simplification-specific productivity effects of AI.

[19] We use Gartner's definition which can be found here.

[20] For definitions of these technologies, we again follow Gartner: see here and here.

[21] To do so, we use the distribution of workers' skills in each occupation implied by the model's pre-AI equilibrium.

Automation. We follow Eloundou et al. (2024) by eliciting automatability by task from large language models. For each of the tasks in O*NET, we ask whether AI can complete it autonomously (i.e., whether task τ is in the automatable set $\mathcal{A}_j$). Eloundou et al. (2024) classify tasks as having either "no", "low", "moderate", "high" or "full" exposure to automation.[22] We classify a task as "automatable" if it has high or full automation exposure, which is restricted to cases where the LLM indicates that generative AI can complete at least 90 percent of the components of the tasks. 22.2 percent of all tasks are classified as automatable by generative AI (see Table B.3). The prompt is documented in Appendix D.2.2.

We find high agreement rates between our measures and those obtained by Eloundou et al. (2024). The share of tasks that are automatable is almost identical across the measures. Importantly for our exercise, we find strong agreement on the share of automated tasks by occupation ($\rho = 0.82$, see Figure A.5b). Table B.2 shows that agreement is also strong on the task-level.

Simplification. Lastly, we elicit the degree to which AI changes tasks' skill requirements, $\boldsymbol{r}_\tau$. In addition to our new data on pre-AI task-level skill requirements, we prompt GPT-4o to evaluate the task's skill requirements before and after workers gain access to generative AI (similar to Hosseini and Lichtinger, 2026). The prompt can be found in Appendix D.2.3. We estimate that across all tasks and skill dimensions, a one-step reduction (out of seven) is the most common change (see Table B.3).

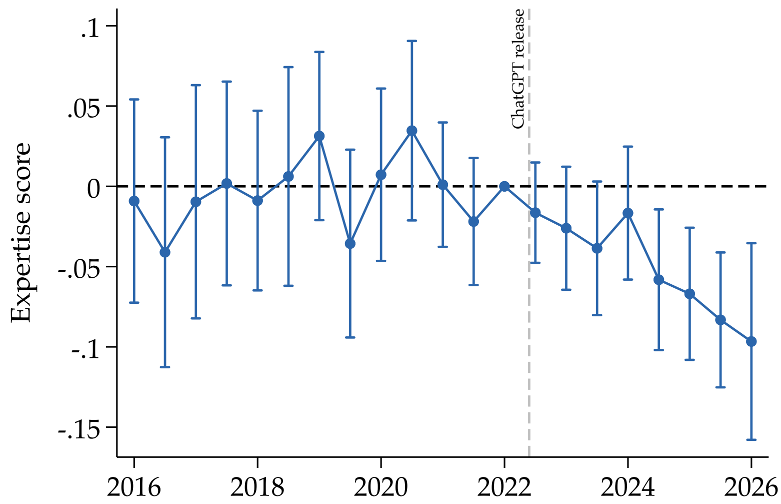
There is large heterogeneity in what AI can simplify across skills: the strongest simplification occurs in time management, writing, judgment and decision making, and critical thinking, versus the least simplification in the manual skills of equipment maintenance, repairing, and installation.

We validate the accuracy of these predicted changes in skill requirements via real-time job postings data. Following Autor and Thompson (2025), we compute expertise indices by occupation over time based on the relative frequency of words in the job postings that are specific to a professional domain. Figure 1 shows that occupations predicted to be more simplified by AI exhibit no differential trend in required expertise prior to ChatGPT's release, but a sharp and accelerating decline thereafter—reaching roughly 0.1 standard deviations by early 2026. This pattern is corroborated by two additional measures of oc-

[22] The specific prompt is documented in (Eloundou et al., 2024, Supplementary Materials).

cupational skill requirements in Figure A.6a: both the frequency of skill-related language and the rate of technical keywords in job postings fall post-2022 in occupations our model predicts to be more simplified.

FIGURE 1: SIMPLIFICATION VIA JOB POSTINGS



Notes: This figure shows that occupations predicted to be more simplified by AI exhibit a decline in required expertise in real-time job postings. Continuous treatment exposure is measured using our occupation-level predicted AI-led simplification. Following Autor and Thompson (2025), the outcome is an expertise index based on the relative frequency of words that are highly specific to a professional domain. We use a random subsample of US job postings in Revelio Labs' Cosmos database. Each coefficient gives the differential effect of a 1 standard deviation higher predicted simplification in t relative to 2022-H1. We include occupation and time fixed effects; standard errors are clustered at the occupation level.

Lastly, simplification aligns with experimental evidence that has consistently shown larger AI-led productivity effects among lower-skilled workers across several occupations (e.g., Dell'Acqua et al., 2023; Noy and Zhang, 2023; Brynjolfsson et al., 2025b; Cruces et al., 2026).

4 Estimation

We jointly estimate the parameters governing productivity, skill accumulation, occupational choices, and the initial skill distribution using maximum likelihood on the NLSY79 estimation sample. We provide a computationally efficient methodology to do so using direct inference. The algorithm recovers the equilibrium prices directly, avoiding the need to solve for the equilibrium within the estimation loop. Table B.5 shows an overview of all model parameters and their estimated values.

4.1 Estimation strategy

The goal of the estimation strategy is to find the parameters that maximize the likelihood of the observed wages and occupational choices. A full maximum-likelihood approach over the complete parameter space is computationally prohibitive given the dynamic and high-dimensional nature of the model. We reduce the computational burden in two main ways.

First, we maximize the likelihood using a sequential approach. We partition the large parameter space into an outer vector $\boldsymbol{\theta}_1$ and a set of objects that can be consistently estimated using a fast procedure for any given $\boldsymbol{\theta}_1$: the vector of task-level productivity parameters $\boldsymbol{\theta}_y = \{\eta, \{\omega_s\}_{s\in S}\}$, occupational amenities $\boldsymbol{\mu} = \{\mu_j\}_{j=1}^J$, and equilibrium prices $\boldsymbol{p} = \{p_j\}_{j=1}^J$. We then maximize the likelihood

$$L\left(\boldsymbol{\theta}_1, \boldsymbol{\theta}_y(\boldsymbol{\theta}_1), \boldsymbol{p}(\boldsymbol{\theta}_1), \boldsymbol{\mu}(\boldsymbol{\theta}_1)\right)$$

with respect to $\boldsymbol{\theta}_1$ only, obtaining consistent estimates of $\boldsymbol{\theta}_y(\boldsymbol{\theta}_1)$, $\boldsymbol{p}(\boldsymbol{\theta}_1)$, and $\boldsymbol{\mu}(\boldsymbol{\theta}_1)$ via closed-form expressions and a fast iterative algorithm. This reduces the dimensions of the nonlinear search by 95 percent.

Second, we only maximize the likelihood of the occupational choices for workers at the terminal age A. Since the occupational choice problem for these workers is static, this avoids repeated solution of the dynamic value function.

Inner algorithm. The inner algorithm proceeds in three steps: infer workers' skills, estimate the task-level productivity parameters using linear regression, and recover occupational amenities from occupational choices.

The first step in the inner algorithm is to compute the workers' skills given $\boldsymbol{\theta}_1$. The NLSY79 provides multi-dimensional skill scores. However, we observe those skills i) only as percentile scores, not as cardinal measures, and ii) only at labor market entry, not later. We first map percentile scores into cardinal skills $\boldsymbol{h}$ using the marginal distribution, approximated with a Beta distribution with parameters included in the outer vector $\boldsymbol{\theta}_1$.[23] That is, we recover the cardinal skills as $\boldsymbol{h} = F^{-1}(\tilde{\boldsymbol{h}})$, where $F(\cdot)$ is the estimated Beta distribution of initial skills

[23]The Beta distribution is a flexible distribution characterized by two parameters B_a and B_b with support on [0,1]. We assume that this distribution is common across skill dimensions.

and $\tilde{\boldsymbol{h}}$ are the workers' percentile skill scores. We then successively apply the skill accumulation function $g_j(\cdot, \psi)$ in equation (16) to infer workers' skills at later ages. That is, given worker i's occupational history $\boldsymbol{j}_i^{a-1} \equiv \{j_{i,1}, \ldots, j_{i,a-1}\}$, worker i's skill level at age a is

$$\boldsymbol{h}_{i,a}(\lambda(\psi_i), \delta) \equiv \left(g_{j_{i,a-1}}(\cdot, \psi_i) \circ g_{j_{i,a-2}}(\cdot, \psi_i) \circ \ldots \circ g_{j_{i,1}}(\cdot, \psi_i) \right) (\boldsymbol{h}_{i,1}) \,.$$

where $(f \circ g)(x) \equiv f\,(g(x))$.[24] Following Heckman et al. (1998), we proxy ψ_i by the Armed Forces Qualification Test (AFQT) score. The parameters $\{\lambda(\psi)\}_{\psi=1}^{4}$ and δ are in $\boldsymbol{\theta}_1$.

We then estimate the task-level productivity parameters and equilibrium prices using a linear regression. The derived occupational wage function in equation (15) governs how skills translate into earnings in each occupation depending on the occupation's price and tasks, and the parameters of the production function $\{\omega_s\}_{s \in S}$ and η. A log-linearization of this function around zero mismatch yields that

$$\log w_j(\boldsymbol{h}_i) \approx \log p_j + \sum_{s \in S} w_s \log(h_{is}) - \eta \sum_{s \in S} \sum_{\tau \in \mathcal{T}_j} \tilde{\theta}_{j,\tau} \min\left\{h_{is} - r_{\tau,s}, 0\right\}^2 \tag{17}$$

where $\tilde{\theta}_{j,\tau} \equiv \theta_{j,\tau} \gamma_\tau^{\rho-1}$ (see Appendix C.2 for the proof).[25] Equation (17) implies that we can estimate the equilibrium prices p_j and the parameters of the production function using a simple OLS regression of wages on occupational fixed effects, skills, and skill mismatch. Since the equilibrium prices can be recovered from the occupational fixed effects in the regression, no equilibrium solution is required within the estimation loop.

We correct for selection on the occupation-specific productivity shocks using the control-function approach of Dubin and McFadden (1984). Observed wages equal the deterministic component $w_j(\boldsymbol{h})$ times the stochastic shock ε_j, so selection arises because workers observe ε_j when choosing their occupation. The Gumbel distribution of $\log \varepsilon_j$ implies that its expectation conditional on choosing occupation j is $\mathbb{E}[\log \varepsilon_j \mid j, \boldsymbol{h}, \psi, k] = -\zeta \log \mathbb{P}_a(j \mid \boldsymbol{h}, \psi, k)$ where $\mathbb{P}_a(j \mid \cdot)$ refers to the probability of choosing j (see e.g., Dubin and McFadden, 1984).[26]

[24] To save notation, it is left implicit above that $\boldsymbol{h}_{i,a}$ depends on $\lambda(\psi_i)$ and δ through $g_j(\cdot, \psi_i)$.

[25] For estimation we set $\sum_{\tau \in \mathcal{T}_j} \tilde{\theta}_{j,\tau} = 1$ for all j and use O*NET's task-importance weights to proxy $\tilde{\theta}_{j,\tau}$. Since we estimate the model on pre-AI data, no task is automated, i.e., $\mathcal{A}_j = \varnothing$.

[26] While Dubin and McFadden (1984) derives this selection equation for a static choice problem, it extends directly to dynamic choice settings like ours.

We thus control for this selection term in the regression.[27]

In the last step of the inner algorithm, we estimate the occupational amenities $\boldsymbol{\mu}$ using a fast iterative procedure to match the occupational choices of terminal-age workers. At age $a = A$, equation (10) simplifies to

$$\mathbb{P}_A(j \mid \boldsymbol{h}, \psi, k) = \frac{\exp\left(\frac{1}{\zeta}\left(\log w_j(\boldsymbol{h}) + \mu_j - \kappa(k,j)\right)\right)}{\sum_{l=1}^{J} \exp\left(\frac{1}{\zeta}\left(\log w_l(\boldsymbol{h}) + \mu_l - \kappa(k,l)\right)\right)}. \tag{18}$$

where ζ (scale of productivity shocks), κ (switching costs) are included in outer vector $\boldsymbol{\theta}_1$ and thus taken as given in this step. The likelihood is maximized with respect to $\boldsymbol{\mu}$ when the observed share of terminal-age workers in occupation j, denoted s_j, equals the model-implied share $\tilde{s}_j(\boldsymbol{\mu})$. We solve for this $\boldsymbol{\mu}$ using the contraction mapping proposed by Berry et al. (1995):

$$\mu_j^{(r+1)} = \mu_j^{(r)} + \Psi\left(\ln(s_j) - \ln\left(\tilde{s}_j(\boldsymbol{\mu})\right)\right) \qquad \text{for some } \Psi \in (0,1] \tag{19}$$

where Ψ is a damping parameter.[28]

Outer algorithm. In the outer algorithm, we optimize over the outer vector $\boldsymbol{\theta}_1$—the parameters governing skill accumulation function, the initial skill distribution, and occupational choice. We choose these parameters to maximize the joint likelihood of the wage function and the occupational choices of workers at the terminal age A. That is,

$$\begin{aligned}\hat{\boldsymbol{\theta}}_1 = \arg\max_{\boldsymbol{\theta}_1} \sum_{i=1}^{N}\sum_{a=1}^{A}\sum_{j=1}^{J} 1[j_{i,a} = j]\ \log \pi\left(\log w_{i,a,j} - \log \tilde{w}_{i,a,j}(\boldsymbol{\theta}_1)\right)\ + \\ \sum_{i=1}^{N}\sum_{j=1}^{J} 1[j_{i,A} = j]\ \log \mathbb{P}_A(j \mid \boldsymbol{h}_{i,A}, \psi_i, k_i; \boldsymbol{\theta}_1)\end{aligned} \tag{20}$$

where $\tilde{w}_{i,a,j}(\boldsymbol{\theta}_1)$ is the expected wage based on the inner-step given $\boldsymbol{\theta}_1$ and $\pi(\cdot)$. The first term captures the likelihood of the observed wages and the second term reflects that of the observed occupational choices. In practice, wages contain noise that workers do not act on—such as measurement error or idiosyncratic pay—which the structural model's choice-relevant shocks $\log \varepsilon_j$

[27] We estimate these probabilities using occupation-specific logit regressions that condition on workers' previous occupation, 10-year age bins, and each dimension of initial skill.

[28] In practice, we use the SQUAREM algorithm to accelerate convergence (Varadhan and Roland, 2008; Reynaerts et al., 2012; Conlon and Gortmaker, 2020).

alone cannot absorb without distorting the occupational choice probabilities. We therefore augment $\log \varepsilon_j$ with an independent Gaussian term $\log \nu_j$ and approximate the total wage shock $\log \varepsilon_j + \log \nu_j$ by a normal distribution $\pi(\cdot)$.[29]

Full-population amenities. As a final step after the outer algorithm has converged, we re-estimate the occupational amenities $\boldsymbol{\mu}$ using the full working-age population rather than the old workers alone. This ensures that the model matches employment shares for all workers. The procedure follows the same BLP iteration, but now solves the worker's full dynamic value function at each iteration.[30] The estimates of the occupational amenities resulting from the inner algorithm and this dynamic BLP procedure are almost identical (correlation: 0.99), validating our computational shortcut.

4.2 Parameter Estimates

Table 1 reports the estimates of the task-level production function parameters in equation (14). The first five columns show the general skill elasticities ω_s, which govern the degree to which higher skills raise productivity across all tasks regardless of those tasks' specific requirements. We find that the returns to math and social skills are highest, consistent with Deming (2017). The mismatch parameter η implies that if a worker's skill falls one unit below a task's requirement in a single dimension (on O*NET's 1–7 scale), their task-specific productivity is approximately 4.4% lower than that in tasks for which they meet the skill requirements in every dimension. This cost of underqualification is substantial and generates meaningful comparative advantage across tasks with differing skill requirements.

[29] The estimated variance of $\log \nu_j$ is substantially larger than that of $\log \varepsilon_j$, so the convolution is dominated by the Gaussian component.

[30] In dynamic problems, this procedure is not generally a contraction mapping and we can thus not prove that the fixed point is unique (see also Gowrisankaran and Rysman, 2012). However, the procedure yields the same results for any starting value we tried.

TABLE 1: PRODUCTION FUNCTION: PARAMETER ESTIMATES

General skill					Mismatch
ω_{Mn}	ω_{Mt}	ω_S	ω_T	ω_V	η
0.330	0.786	0.545	0.220	0.361	0.044
(0.024)	(0.028)	(0.021)	(0.038)	(0.031)	(0.002)

Notes: This table shows estimates of the task-level productivity parameters. Subscripts Mn, Mt, S, T, V refer to manual, math, social, technical, and verbal, respectively. Estimates are obtained through OLS based on equation (17). Standard errors in parentheses (not corrected for uncertainty in other parameters).

The occupational prices are recovered from the occupational fixed effects in the production function regression. Consistent with the model's prediction that occupational prices reflect skill requirements, the estimated prices $\hat{p}_j$ are strongly positively correlated with occupational skill requirements: skill requirements explain around 73% of the variance in prices across occupations (see Table B.6). To reduce noise in these price estimates, we apply empirical Bayes shrinkage toward the price predicted by skill requirements (Walters, 2024).

Table 2 presents the estimates governing initial skills and skill accumulation. The depreciation rate $\delta = 0.0003$ implies that skills are highly persistent when workers only perform tasks for which they are overqualified. The learning cost $\lambda(\psi)$ is strictly decreasing in the AFQT quartile, implying that higher-ability workers converge more quickly toward the skill requirements of tasks for which they are underqualified. The shape parameters of the initial Beta distribution imply an average initial skill level of $B_a/(B_a + B_b) = 0.35$, corresponding to 3.11 on O*NET's 1–7 scale, in between the "low" and "medium" skill requirement anchors. Figure A.8 plots the implied density function.

TABLE 2: SKILLS AND SKILL ACCUMULATION: PARAMETER ESTIMATES

Learning costs				Depr.	Initial dist.	
$\lambda(1)$	$\lambda(2)$	$\lambda(3)$	$\lambda(4)$	δ	B_a	B_b
3.50	2.97	2.81	2.67	0.0003	61.74	113.84

Notes: This table shows parameter estimates for the law of motion for skill accumulation in (16) and the initial skill distribution. $\lambda(\psi)$ refers to the learning cost at quartile ψ of the AFQT distribution.

Lastly, we estimate the scale parameter $\hat{\zeta} = 0.053$ and the switching cost parameter $\hat{\kappa} = 0.340$. The estimate for κ implies that the utility cost of switching occupations is equivalent to a 29% wage loss.

Calibrated parameters. Some parameters are calibrated externally rather than estimated. We set career length A to 40 so that each model period corresponds one year between ages 25 and 64. Following Keane and Wolpin (1997), the discount factor $\beta = 0.78$ (see also Postel-Vinay and Robin (2002) for similar estimates). We set the elasticity of substitution between tasks ρ to 0.49, as estimated by Humlum (2021), implying that tasks within an occupation are complements.[31] The task sets $\mathcal{T}_j$, importance weights $\theta_{j,\tau}$, and the skill requirements are constructed from the data described in sections 3.1.1 and 3.1.2.

The labor share Γ_j in each occupation—the share of occupational revenue accruing to workers—is calibrated from the share of automatable tasks and the cost savings AI generates in those tasks. Appendix C.1 shows that this share satisfies

$$\Gamma_j \approx \frac{\chi^{\rho-1}\left(1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau}\right)}{\sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} + \chi^{\rho-1}\left(1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau}\right)}, \tag{21}$$

where χ is the unit cost of producing automatable tasks with AI relative to the unit cost of producing them with labor. Using experimental evidence, Acemoglu (2025) estimates the cost savings of AI in automatable tasks to be 27%. Hence, we set $\chi = 0.73$.

We close the model by estimating the demand for each occupational good. Our baseline specification assumes that occupational goods are substituted with a constant elasticity of substitution (CES) σ, so that demand for occupation j satisfies $D_j(\boldsymbol{p}) \propto \alpha_j p_j^{-\sigma}$. We set the elasticity of substitution between occupations $\sigma = 1.57$—the midpoint between 1.81 (Burstein et al., 2019) and 1.34 (Caunedo et al., 2023). The CES importance weights $\{\alpha_j\}_{j=1}^{J}$ are then identified from observed occupational wage shares and the estimated occupational prices because for any two occupations i and j,

$$\frac{\alpha_i}{\alpha_j} = \left(\frac{p_i}{p_j}\right)^{\sigma-1} \times \frac{\text{Wage share of occupation } i}{\text{Wage share of occupation } j}.$$

[31] For the nested CES, we set the elasticity of substitution between 1-digit sectors to 1.01 and across 3-digit industries within sectors to 2.98 (as estimated by Hobijn and Nechio, 2019). Occupations can be substituted with an elasticity of 0.90 within an industry (Goos et al., 2014).

We compute occupational wage shares from the 2018 BLS Occupational Employment and Wage Statistics (OEWS) and use the occupational fixed effects from equation (17) as consistent estimates of log occupational prices. Because the supply-side parameters are estimated independently of demand, this demand specification does not affect any of the other parameter estimates.

In addition to this baseline CES structure for occupational demand, we estimate a nested CES structure. Occupations are combined to produce goods for each industry, where industries are characterized by the varying importance weights of occupations. These industries (3-digit NAICS) are nested within sectors (1-digit NAICS). For this nested CES, we set the elasticity of substitution between 1-digit sectors to 1.01, across 3-digit industries within sectors to 2.98 (as estimated by Hobijn and Nechio, 2019), and across occupation within industries to 0.90 (Goos et al., 2014). Each occupation's weight is inferred from its wage bill share within an industry (analogous to the standard CES case). These occupation-level weights are then aggregated to obtain industry- and sector-level importance weights.

4.3 Model Fit

The model's steady state moments fit labor market data well. The model's moments are computed from a simulated sample of 100,000 workers living in the steady state before any technical change occurs. Figure 2 reports how well the moments from this simulated panel match the data.

First, Figure 2a shows that the model captures the unconditional distribution of wages reasonably well. Given that some drivers of wage inequality, such as regional, racial, and gender differences, are omitted from the model, it is not surprising that inequality is somewhat underestimated. However, this underestimation is quite limited. For instance, the ratio between the 75$^{\text{th}}$ and the 25$^{\text{th}}$ percentile is 2.04 in the data, compared to 1.84 in the model and the top 10% wage share is 20% in the model, compared to 26% in the data. Table B.7 reports how additional inequality measures compare between the model and the data.

The model also accurately replicates patterns of occupational sorting. Figure 2b shows the correlation between the average skill by occupation in the model and the NLSY79 data. We compute these correlations, using only young workers for which we observe the skills directly from ASVAB scores (rather than partially inferred through the skill accumulation function), making the test as stringent as possible. The correlations range between 0.6 and 0.8 across skill dimensions, implying that the skill assessment scores in the NLSY79 are predictive of occupational choices (see also Lise and Postel-Vinay, 2020) and that workers in the model select into occupations based on their skills in ways similar to that observed in the NLSY79.

Figure A.9a shows that the average wage by occupation matches the data almost perfectly. This is the most directly targeted moment, as we estimated demand based on occupational wage shares and job-specific amenities based on occupational employment shares.

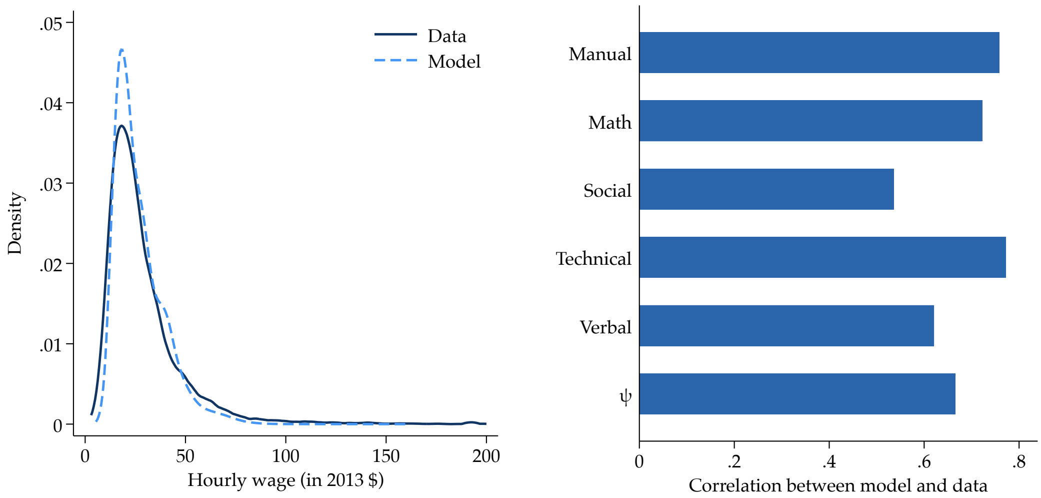
The median wage by age also matches the pattern observed in the data. Figure A.9b shows that the model matches the growth rate of wages from labor market entry to around age 55. However, the wage pattern in the model is not as concave as in the data, so that growth in the first years is underestimated and growth in the last 10 years overestimated. Furthermore, the model predicts markedly higher wages in the first period than those directly after. This feature is caused by the fact that occupational switching costs are only incurred after the first period. Workers are therefore more likely to choose occupations in which they are highly productive (i.e., with a high ε_j) in the first period than in any later periods.

The model also accurately matches the observed degree of occupational persistence. The probability of remaining in the same 3-digit occupation is 0.86 in the model and 0.90 in the CPS data. This moment is directly targeted by the switching cost parameter κ. The model additionally fits the untargeted probability that a worker stays within a broader 2-digit occupational group: 0.92 (model) and 0.94 (CPS data). In other words, even though the occupational switching cost applies equally across all but one occupation, the model captures that workers are more likely to stay within a similar set of occupations.

We further compare the transition probabilities between occupations conditional on switching. The correlation between the (log of) the transition probabilities in the model and data is 0.56 on the 2-digit occupation level. On the 3-digit level, it is substantially lower: 0.20. In other words, the model accurately predicts occupational transitions across 23 broader occupational groups. Within those groups, occupational transitions are harder to predict, because occupations are more similar in skill requirements within those groups.

FIGURE 2: MODEL FIT: COMPARING MODEL MOMENTS WITH DATA

(A) WAGE DISTRIBUTION (B) SKILL SORTING BY OCCUPATION



Notes: Panel A shows a kernel density plot of the wage distributions of the NLSY79 data and the model's steady state. Panel B reports the correlation between the average skill by occupation in the model's first period and the NLSY79.

5 Artificial Intelligence and the Labor Market

Having estimated the model and validated its ability to match various features of the pre-AI labor market, we next use the model to assess counterfactual scenarios. Specifically, we study AI's general equilibrium effects on wages, inequality, welfare, skill returns, and occupations using our predictions of the technology's capabilities to augment, automate, and simplify.

5.1 AI's Effect on Wages and Inequality

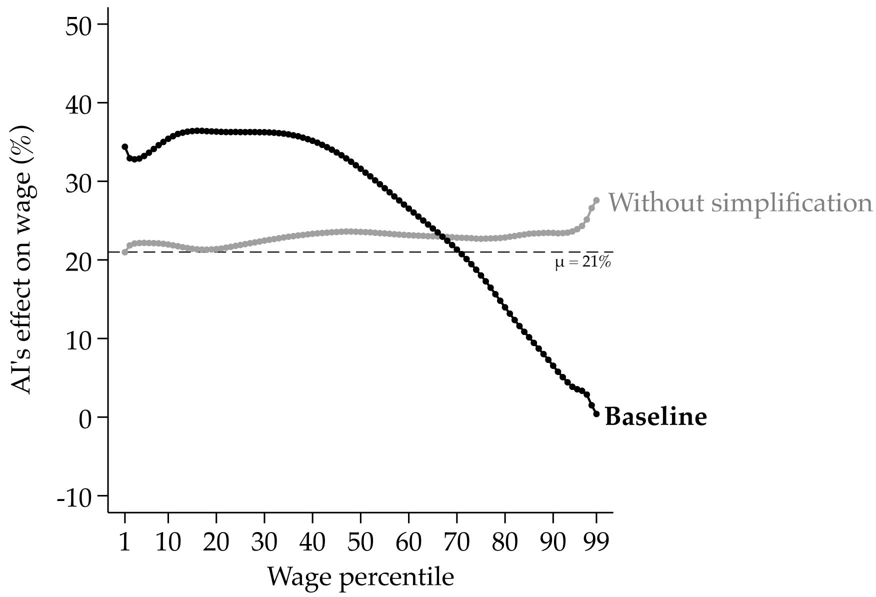
We begin by studying AI's impact on the steady state wage distribution. We find that AI produces sizable average wage gains of 21 percent (see Figure 3). This effect is slightly larger than estimates of IT's cumulative growth contribution of around 16 percent (Jorgenson et al., 2008). It is also similar to other estimates on AI's growth effect (Baily et al., 2023; Aghion and Bunel, 2024; Filippucci et al., 2024) but larger than the estimate by Acemoglu (2025).[32]

[32]The productivity effects quantified by Acemoglu (2025) are based on a conservative automation scenario, whereas our evidence also incorporates augmentation.

The three mechanisms of technical change shape wages through competing forces (see Figure A.11). Augmentation has an unambiguously positive effect on wages, and we find it to be quantitatively large (18% on average). Automation raises productivity but displaces labor with capital (Acemoglu and Restrepo, 2018); here the productivity effect dominates, raising average wages by 6%.[33] Simplification weakly increases productivity (see equation 14), but limits opportunities for learning; for AI, we find that the net effect is slightly negative (−3%). This learning cost is consistent with experimental evidence that AI usage slows the acquisition of new skills (Shen and Tamkin, 2026).

The wage gains are concentrated at the bottom of the distribution and are nearly zero at the 99th percentile, implying a significant drop in wage inequality.

FIGURE 3: WAGE EFFECTS ACROSS THE DISTRIBUTION



Notes: This figure shows the distribution of wage changes induced by generative AI across the wage percentile distribution. The horizontal axis represents wage percentiles weighted by pre-AI employment, and the vertical axis shows the percentage change in wages for each percentile. The black line shows the joint effect of AI's augmentation, automation, and simplification on each wage percentile. The gray line shows the effects if AI induced only augmentation and automation, but no simplification.

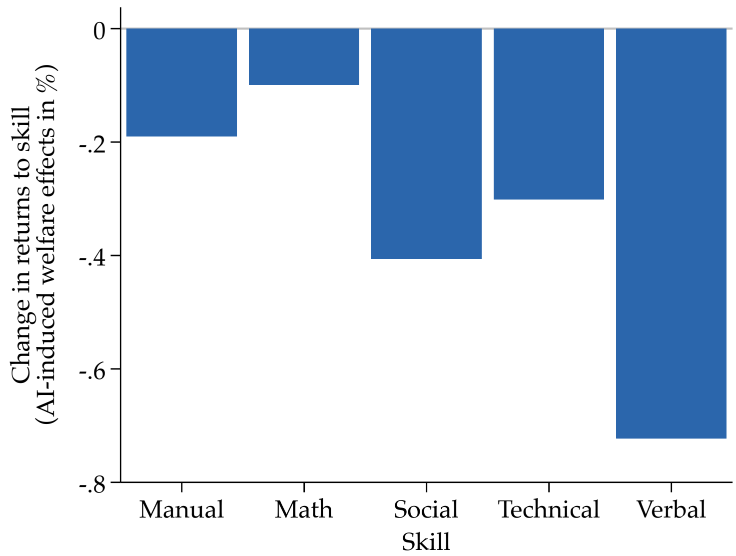
Simplification is the key force lowering inequality. Without simplification—that is, with only augmentation and automation operating—inequality would slightly *increase* rather than strongly *decrease*. Simplification lowers inequality in two ways. First, it reduces wage dispersion within occupations by enabling

[33]This finding is primarily driven by the cost savings of automation, which, following Acemoglu (2025), we calibrate to 27%.

lower-skilled workers to perform tasks more productively. Second, it reduces wage differences across occupations by making occupations with high skill requirements more accessible to lower-skill workers, reducing its relative price.

In contrast, automation and augmentation have small distributional effects. First, within occupations, augmentation and automation affect the relative productivity of workers with different skills only if they disproportionately affect tasks with atypical skill requirements. In such cases, augmentation and automation induce an indirect form of simplification by changing tasks' effective importance within an occupation. However, we find that these indirect effects are quantitatively small relative to direct simplification. Second, across occupations, augmentation and automation can in principle increase wage inequality by increasing the relative productivity of high-wage occupations. However, when the elasticity of substitution across occupations is close to one—as empirical estimates suggest (e.g., Burstein et al., 2019; Caunedo et al., 2023)—such productivity differences translate only weakly into relative wage changes.

FIGURE 4: HOW AI CHANGES THE RETURNS TO SKILLS



Notes: This figure shows how AI's welfare effects differ by skills. The welfare effects are measured in equivalent permanent percentage wage increases. This figure plots the coefficient of a regression of these welfare effects on skill levels across all dimensions. For interpretability, the skills are expressed on the O*NET scale from 1 to 7.

Consistent with the decrease in wage inequality, we find that the ex-ante welfare gains are largest for lower-skill workers. Specifically, we compare expected welfare at labor market entry, conditional on initial skills, in economies with and without AI. We measure welfare changes using equivalent variation, measured in terms of a permanent proportional wage increase that delivers the same welfare gain as the introduction of AI (regardless of skill and occupation). Figure 4 shows the coefficients of a regression of the welfare gains on initial skill levels. Workers with high verbal skills see the smallest increases in welfare gain: a 1-point increase in verbal skills (on the O*NET scale from 1 to 7) decreases the welfare gains from AI by 0.7%. Higher math skills, in contrast, have the least negative association with AI-induced welfare effects.

Unlike the skill-biased technical changes of recent decades, which increased the relative demand for cognitive skills and widened wage differentials (Katz and Murphy, 1992; Autor et al., 2008), AI's distributional effects are dominated by simplification, which compresses rather than widens the wage distribution. Simplification is similar in spirit to "nonautonomous AI", which allows lower-skill workers to solve difficult problems with AI, and thus raises their wages relative to skilled workers (Ide and Talamàs, 2025).

The decline in the returns to verbal and social skills has implications beyond overall inequality. Hurst et al. (2024) show that rising returns to abstract, cognitive tasks—closely related to verbal and social skills—together with discrimination of Black workers explain why the Black-White wage gap stopped closing after 1980. Our finding that AI compresses these returns suggests that simplification could partially reverse this force.

Our key result is not sensitive to the structure of demand for occupations. Comparing standard and nested CES demand, we find that the distributional impacts are almost identical (see Appendix A.10). On average, wages increase slightly more with the nested versus the standard CES.

5.2 Short- and Long-Run Welfare Implications

While wages are a natural measure of AI's labor market impact, they capture only part of workers' overall welfare. First, wages abstract from the switching cost workers bear to reach the new equilibrium. Second, even if AI raises average wages, its distributional consequences matter: under standard concavity of utility, wage gains accruing to lower-income workers generate larger welfare improvements than equivalent gains at the top of the distribution.

We first consider AI's welfare effects in the short run. Specifically, we consider a one-off ("MIT") shock through which AI is introduced to the economy.

We measure welfare effects in terms of equivalent wage variation. That is, we ask: Absent AI, how much would a worker have to be paid more for the rest of their life to make them equally well off as they are in the transition to AI.

We find that welfare increases despite the reallocation costs that workers face during the transition to AI—for most of them substantially (see Figure A.13). Perhaps unsurprisingly, AI affects older workers' welfare particularly heterogeneously (see Figure A.14). Older workers have already invested in occupations and skills and have less time left to benefit from costly adjustments. Thus, those who have invested in the "wrong" occupations and skills gain less than others.

We also consider scenarios in which AI does not provide any augmentation, automation, or simplification. We find that a small set of workers experience negative welfare effects in the scenario where AI does not augment the worker at all (see Figure A.15 for the distribution under each scenario). However, even in that relatively pessimistic scenario the average worker experiences an 11% welfare gain. In all other scenarios, we find positive welfare effects across the board.

Last, we consider the welfare effects in the long run for workers entering the labor market in the new steady state.[34] We find that long-run welfare increases by 31 percent on average, ranging between 26 and 34 percent for most workers (see Figure A.12). The welfare gains exceed the average wage increase (21 percent) because of reduced inequality: AI disproportionately raises wages at the bottom of the distribution, where the marginal utility of consumption is highest. The average welfare gain is higher in the long run than in the short run because the transition requires costly reallocation. The most important difference between the short and long run is that welfare gains are more unequally distributed in the short run.

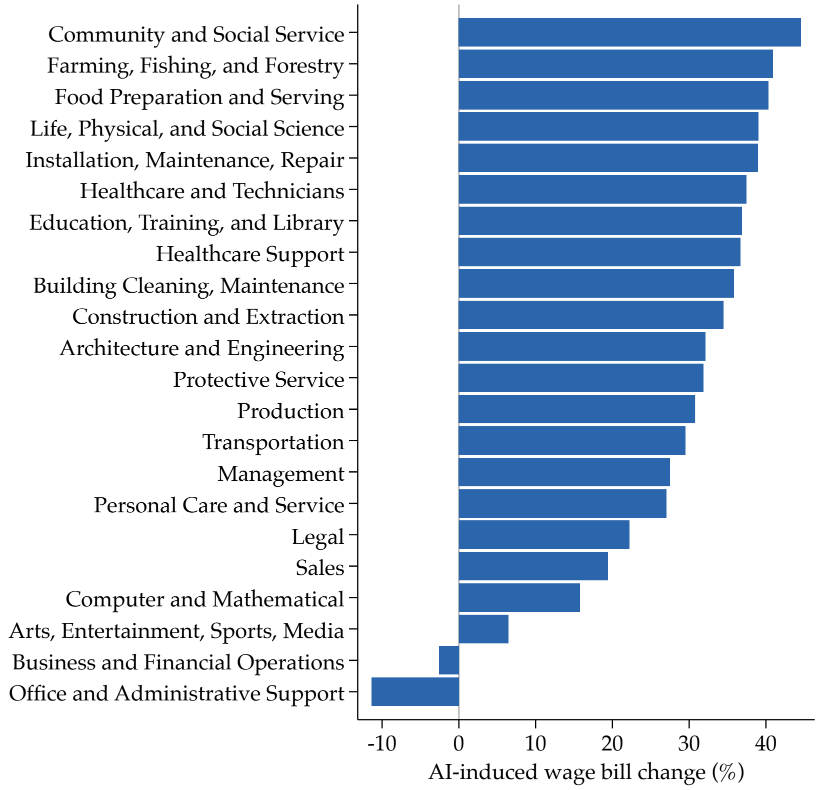
5.3 How Do Occupations Change?

Workers' response to AI significantly shifts the occupational landscape. These shifts are reflected in the changing importance of individual occupations as measured by their wage bill (employment times wages). *Community and Social Service* experiences the largest wage bill increase of over 40 percent, while *Office and Administrative Support* sees an absolute decline in its wage bill by over

[34]The welfare gain is expressed in equivalent wage variation as explained above.

10 percent (see Figure 5).

FIGURE 5: AI'S EFFECT ON OCCUPATIONS' WAGE BILLS



Notes: This figure shows the model predictions on AI's wage bill effects by occupational group. Figure A.18 shows these effects on the disaggregated occupations used in the model.

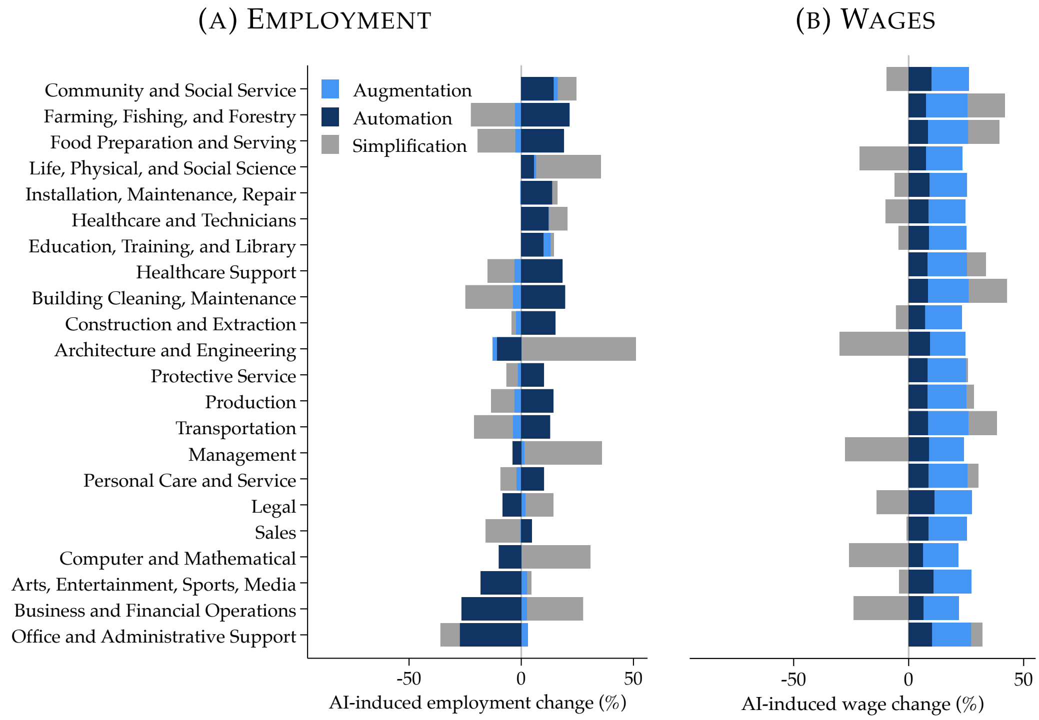
Occupations' wage bill changes are driven by both wages and employment (see Figure A.16). Although wages rise on average, some occupations experience absolute wage declines. The effects on employment and wages often work in the opposite direction (Figure A.17). For instance, *Architecture and Engineering* experiences the largest increase in employment share and the largest decrease in average wages. In contrast, *Building Cleaning and Maintenance* experiences the largest increase in average wages and a decline in employment. Note that because of workers' reallocation, occupation-level wage declines do not imply that individual workers experience wage losses. Since some occupations lose more than half of their employment, considering worker reallocation is key to understand how AI affects workers (Figure A.16c).

We find that simplification shapes relative wage effects more than augmentation and automation. We decompose occupational outcomes by recomputing them under all possible channel combinations (Shapley, 1953; Owen, 1977, see Figure 6). Augmentation and automation tend to increase the productivity of all workers in an occupation equally, unless the skill requirements of affected tasks greatly differ from the occupational average. Small automation- and augmentation-driven relative wage effects imply that such *indirect* simplification has limited quantitative importance for AI.

In contrast to augmentation and automation, *direct* simplification has strong effects on relative wages. Simplification expands the pool of workers who can perform an occupation productively by lowering skill requirements. It thereby raises affected occupations' employment and compresses their average wages through increased competition. Figure A.19 shows this breakdown for more detailed occupational categories.

What characterizes occupations that gain the most from AI? There is a negative relationship between occupations' wage bill gains and their pre-AI skill requirements and education levels (see Figure A.21a). While the patterns are not monotonic, occupations that required the highest skill levels before AI are also the ones with the lowest wage bill gains. This finding holds across all skill dimensions as well as workers' pre-AI college share.

FIGURE 6: AI'S EFFECT ON OCCUPATIONAL EMPLOYMENT & WAGES



Notes: This figure shows the model's predictions on AI's employment and wage effects by occupational group. Occupations are sorted in descending order of AI's effect on the total wage bill, so that the first listed occupation experiences the largest wage bill increase. We conduct a Shapley-Owen decomposition to separate the overall change into the contribution of each channel: augmentation, automation, and simplification.

5.4 AI with Physical Capabilities

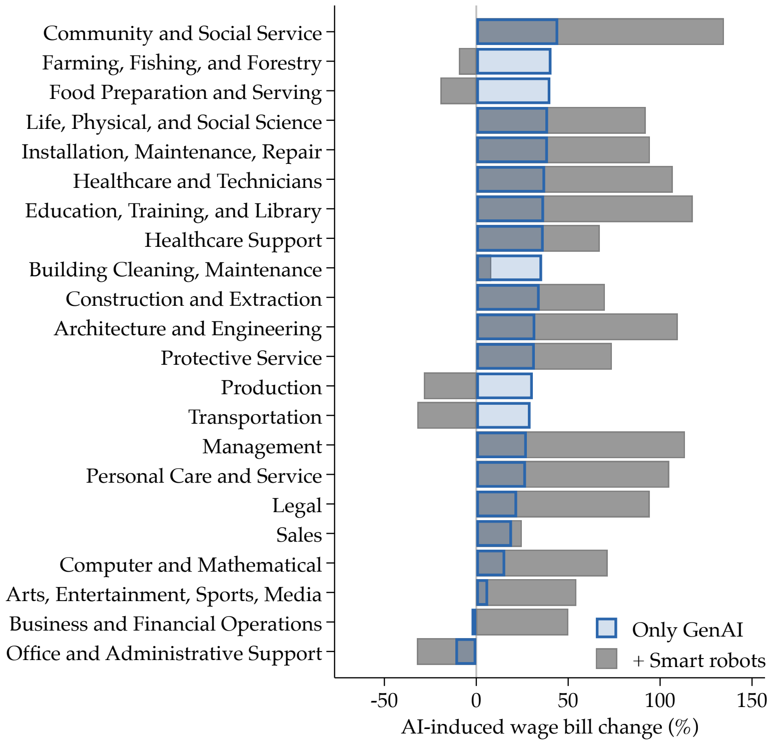
We also examine how our labor market predictions change when considering AI technologies that combine generative and physical capabilities in the form of AI-powered robots. These estimates are necessarily subject to more uncertainty, but highlight how technological capabilities determine which human skills retain value and thus which workers benefit or lose.

The addition of physical capabilities substantially amplifies AI's labor market impact on the wage distribution. Average wages rise by 39 percent in this scenario (compared to 21 percent with only generative AI). The combined effects of physical and generative AI on wage inequality are similar to the effects of generative AI alone. We have previously shown that inequality decreases due to generative AI's capability to simplify work, whereas we model smart robots as only expanding AI's automation and augmentation capabilities—channels which we find have smaller effects on inequality.

There are notable shifts in the patterns of occupational reallocation as AI gains physical capabilities (see Figure 7). *Community and Social Service* and *Education, Training, and Library* occupations are the main winners, more than doubling in wage bill. In contrast, a larger number of occupational groups now lose over a quarter of their wage bill. The most striking reversals occur for occupations in *Food Preparation and Serving*, *Production*, *Transportation*, and *Farming, Fishing, and Forestry*. Those are occupations requiring manual tasks that AI with physical capabilities (but not generative AI) can automate (see also Figure A.22a).

Relative to generative AI, the addition of physical capabilities further decreases the returns to social, technical, and verbal skills, but not the returns to manual and math skills.

FIGURE 7: AI'S IMPACT WITH VS. WITHOUT PHYSICAL CAPABILITIES



Notes: This figure shows the model's predictions on AI's wage bill effects by occupational group under two scenarios: one with only generative AI and one where AI systems also possess physical capabilities. Occupations are sorted in descending order of generative AI's effect on the total wage bill, so that the first listed occupation experiences the largest wage bill increase in the generative AI scenario.

6 Early Signs of AI's Labor Market Impact

In this section, we compare our model's theoretical predictions to empirical evidence. First, we use recent labor market data from the CPS to test whether the model's predictions on the labor market effects of generative AI are beginning to unfold. Specifically, our event study assesses whether predicted occupational outcomes correlate with observed changes since the release of increasingly capable LLMs starting in late 2022. Second, we relate predicted changes in returns to skills to predicted changes in college major choices and implement a similar event study using data from the National Student Clearinghouse. Third, we zero in on three occupations frequently discussed in relation to AI's impact to explain and validate the model's predictions.

6.1 Event Study of Labor Market Shifts

We implement an event study design using Current Population Survey (CPS) data from 2020 to 2025 (Flood et al., 2025). Our continuous treatment variable is each occupation's model-predicted change in occupational outcome $\Delta\tilde{Y}_j$ (wage bill share, employment, or wages). The specification assesses whether the model predicts differential occupational trends after the release of increasingly capable LLMs starting in late 2022, conditional on occupation fixed effects (α_j), time fixed effects (γ_t), and time fixed effects interacted with occupation-level controls ($\eta_t \cdot X_j$):

$$Y_{j,t} = \sum_{k \neq -1} \beta_k \cdot \mathbb{1}[t = k] \times \Delta\tilde{Y}_j + \alpha_j + \gamma_t + \eta_t \cdot X_j + \epsilon_{j,t}. \tag{22}$$

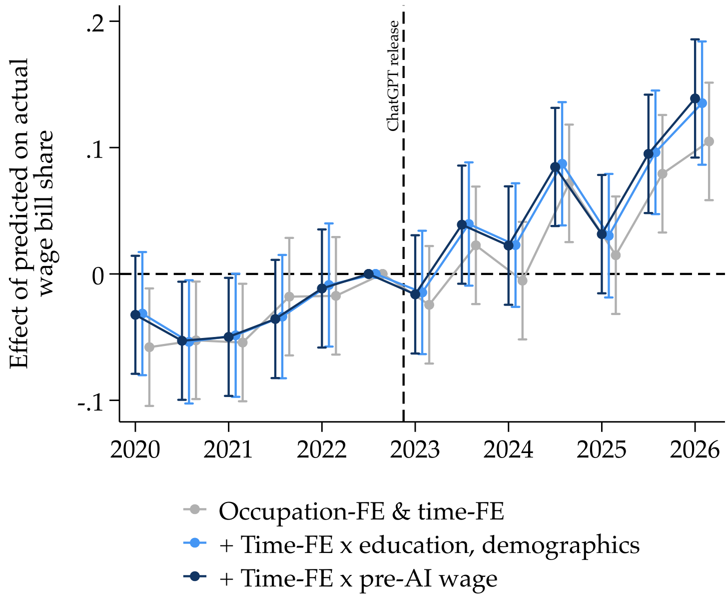
The coefficients β_k capture differential trends for occupations with higher predicted change, with $\beta_k = 1$ indicating that the full model-predicted effect has materialized by period k.[35]

Figure 8 presents our event study estimates. Occupations predicted to gain importance based on their wage bill share indeed begin to see a relative increase starting around two years after OpenAI's first release of ChatGPT. This effect gradually increases over time. The magnitude of our estimates suggests that by early 2026, 13 percent of the predicted wage bill share gains had materialized.

[35]To reduce sampling noise, we aggregate monthly CPS data into 6-month periods to compute the occupational outcomes $Y_{o,t}$. Our sample includes working-age individuals (18-65) in the labor force.

Figure A.26 shows results for the wage bill's two components: employment and wages. Employment begins to rise significantly for occupations predicted to gain employment from AI starting around one year after ChatGPT's release. In contrast, we do not observe any meaningful effects on wages, suggesting that the initial adjustment of the labor market occurs mostly through quantities not prices (this finding is consistent with evidence on young workers from Brynjolfsson et al., 2025a).

FIGURE 8: EARLY LABOR MARKET EFFECTS OF GENERATIVE AI



Notes: Event study estimates ($\widehat{\beta}_k$) show differential changes in occupational wage bill shares following ChatGPT's November 2022 release. Coefficients represent the effect of the model-predicted change in the outcome on the observed outcome. A coefficient of 1 would indicate complete realization of model predictions. Estimates use CPS data aggregated to 6-month periods with occupation fixed effects and time fixed effects. The specification with controls includes occupation-specific trends based on education, sectoral composition, demographics, and pre-AI wages. Error bars represent 95%-confidence intervals.

We interpret this evidence as suggestive, particularly given that some employment effects appear to predate 2022. While this timing could indicate that other factors correlated with AI exposure are driving these patterns, an alternative explanation is that occupations most exposed to AI (such as radiologists and telemarketers) had already begun adopting generative AI tools before ChatGPT's release (see, e.g., Acemoglu et al., 2022; Goldsmith-Pinkham et al., 2026).

6.2 Event Study of Shifts in College Major Choices

We next assess how the returns to majors change in the era of AI and whether college students have already begun adjusting their major choices. Major choices offer useful insights on early responses to technical change as young, higher-educated workers are among the earliest AI adopters (Bick et al., 2024; Hartley et al., 2024; Chatterji et al., 2025; Humlum and Vestergaard, 2025) and major choices can adjust quickly. In contrast, labor market adjustments require occupational switching and retraining.

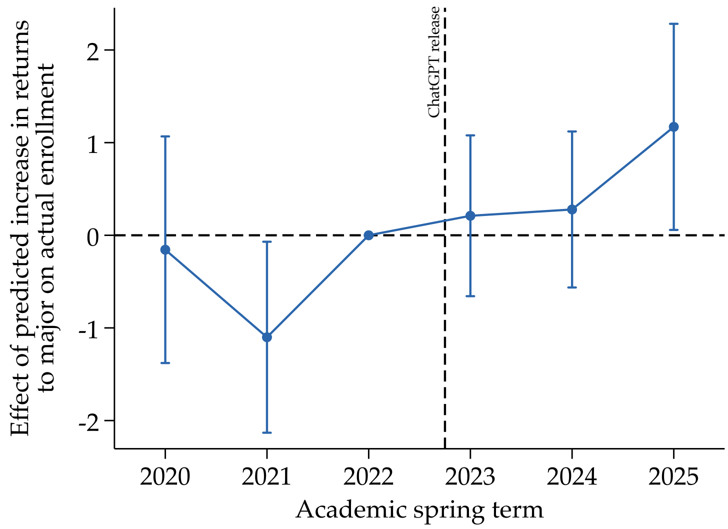
To construct predictions for the changes in returns to each major, we combine our model-based estimates on AI-induced changes in the labor market returns to skills with skill intensity measures from the Course-Skill Atlas, which maps majors to O*NET tasks via information from course syllabi (Javadian Sabet et al., 2024). We link their major-task mapping to our task-skill framework, yielding skill intensities across our five dimensions for each major. As the main outcome, we use major enrollment from the National Student Clearinghouse, covering 96 percent of all US post-secondary students.

Majors intensive in skills that retain most value under AI—particularly math and manual skills—see the largest predicted gains (see Figure A.23). Mechanics, engineering, architecture, physics, and chemistry rank among the top majors whose returns rise relative to other majors by over two percentage points. In contrast, majors intensive in verbal and social skills, where our model predicts the largest decline in returns, fare worst. Women's studies, political science, religion, social work, and languages appear at the bottom of the distribution.

Using our predictions of each major's changing returns, we implement an event study mirroring our analysis of labor market outcomes. The specification follows equation (22), with college major enrollment as the outcome and predicted changes in returns to each major as the continuous treatment variable.

After 2022, a positive relationship emerges: students increasingly enroll in majors that our model predicts will benefit from AI-induced changes in skill returns (see Figure 9). By the spring of 2025, moving from the bottom to the median major (in terms of AI-induced changes in returns) implies a doubling in enrollment. Before 2022, the coefficients fluctuated around zero with no discernible trend, showing that majors predicted to benefit from AI were not already experiencing differential enrollment growth. This evidence suggests that students are already reallocating to fields better positioned for an AI-transformed labor market.

FIGURE 9: EARLY COLLEGE MAJOR EFFECTS OF GENERATIVE AI



Notes: This figure shows event study estimates ($\widehat{\beta}_k$), corresponding to differential changes in college major enrollment (in logs) following ChatGPT's November 2022 release. Coefficients represent the effect of the model-predicted change in returns to each major on observed log enrollment. A coefficient of 1 would indicate that a one percentage point increase in a major's returns (relative to the average major) is associated with a doubling in enrollment. Estimates use National Student Clearinghouse data with major fixed effects and time fixed effects. Error bars represent 95%-confidence intervals clustered at the major level.

6.3 Case studies

Case studies of three specific occupations provide further understanding of and support for our model's predictions: radiologists, management analysts, and telemarketers.

Radiologists. In 2016, deep learning pioneer Geoffrey Hinton warned to "stop training radiologists," because AI would render them obsolete within five years. Since then, radiology has indeed accounted for more than 75 percent of all FDA-authorized clinical AI tools, and roughly two-thirds of US radiology departments report using AI (Mousa, 2025). However, the labor market for radiologists has grown: the wage bill share increased by 6.6 percent between 2016 and 2024. This increase is driven by strong employment growth (23.2% versus average of 9.8%) and attenuated by below-average wage growth (30.1% versus average of 36.9%).

Our model's predictions line up with these observed labor market patterns. We predict a 42 percent wage bill increase, 1.8 times larger than the average (see Figure A.18). In line with observed occupational changes, the model predicts above-average employment growth (28% versus average of 0% by construction) and below-average wage growth (11% versus average of 21%; see Figure A.19).[36]

Simplification is the key to understanding these outcomes. Radiology experiences strong simplification, pushing employment up and relative wages down. Automation, which affects only a small share of radiologists' tasks relative to other occupations, increases radiologists' employment further (while having small effects on relative wages); augmentation generally does not strongly affect employment or relative wages.

Management Analysts. We predict that the importance of management analysts (management consultants) will shrink. The model predicts a 4 percent decline in employment and no change in wages—a wage effect well below the average increase of 21 percent (see Figure A.19).[37]

The reason for this prediction is that management analysts are strongly exposed to both simplification and automation. Simplification pushes average wages down and employment up. However, automation offsets the positive effects on employment, so that the occupation ends up being negatively affected in both dimensions.

The pattern of simplification in management consultancy aligns with experimental evidence. Dell'Acqua et al. (2023) find that lower-skilled consultants experience larger productivity gains from AI. Through the lens of our model, this can only be explained through (direct or indirect) simplification: the reduction in skill requirements increases the relative productivity of lower-skill workers.

Telemarketers. Telemarketers represent one of the clearest cases where AI substitutes for, rather than complements, human labor. Their work consists of

[36]The model outcomes are for *Health Diagnosing and Treating Practitioners*, the 3-digit occupational group that includes radiologists.

[37]The model outcomes are for *Business Operations Specialists*, the 3-digit occupational group that includes management analysts.

12 distinct tasks, all of which can be automated by generative AI. The broader occupational group, *Other Sales and Related Workers*, also faces high automation exposure, strongly reducing employment. This decline in employment puts telemarketers among the 5 percent of most negatively affected occupations in terms of both employment and total wage bill (see Figures A.18 and A.19). Simplification does not play a large role for this occupation, so that relative wages remain more or less constant.

7 Conclusion

Technical change reorganizes production at the task level, so understanding its labor market effects requires characterizing workers' comparative advantage across occupations and tasks. This paper develops and estimates a dynamic task-based framework that recovers this comparative advantage and embeds it in a general equilibrium model of occupational choice and skill accumulation. We use this framework to study artificial intelligence as a technology that augments, automates, and simplifies tasks. The quantified model predicts that AI substantially raises wages, especially in the lower part of the wage distribution. Simplification of tasks is the key reason why our model predicts that AI will reduce inequality. Recent trends in labor markets and college major choices show that AI's predicted effects have already begun to unfold.

Our work has several limitations that future research may be able to overcome. First, our estimates are based on our expectations (as of 2026) of AI's future capabilities, but those expectations are evolving quickly and there is substantial uncertainty around them. We believe our estimates are most suitable for a 10 year time horizon. Second, our model does not feature unemployment. While we find that AI generates a large reallocation of workers across occupations, we ignore frictions that may generate transitional unemployment accompanying this reallocation. Third, this paper only considers the effect of technical change on wage income. Beyond wages, AI also has the potential to affect inequality through the incomes of entrepreneurs and capital owners. On the one hand, small-scale entrepreneurs specifically may benefit from the low fixed cost of using AI, making them more competitive with large-scale entrepreneurs, lowering inequality. On the other hand, producing AI requires high fixed costs, which can increase firm concentration and raise inequality (Reichardt, 2025). Furthermore, the owners of capital may benefit disproportionately if the returns to capital rise faster than wages (Moll et al., 2022).

Future research may also be able to address several additional dimensions we hold fixed. In our framework, we take the technical change brought about by AI as exogenous. One could, however, consider how simplifying technologies arise from directed innovation when particular skills are in short supply (Acemoglu, 2002; Acemoglu and Restrepo, 2018). Similarly, we hold the organization of work fixed. In practice, technological change may prompt the emergence of new tasks and a reorganization of how work is bundled into occupations. We also abstract from adoption frictions. Though AI adoption appears rapid relative to previous technologies (Bick et al., 2024), differential adoption rates across skill groups could attenuate the equalizing effects of simplification if lower-skilled workers are slower to adopt (Hartley et al., 2024; Chatterji et al., 2025; Humlum and Vestergaard, 2025; Hassan et al., 2026). Last, given that we find empirical evidence for AI's impact on college majors, it is useful to incorporate how such educational choices respond to technical change in the model (Heckman et al., 1998).

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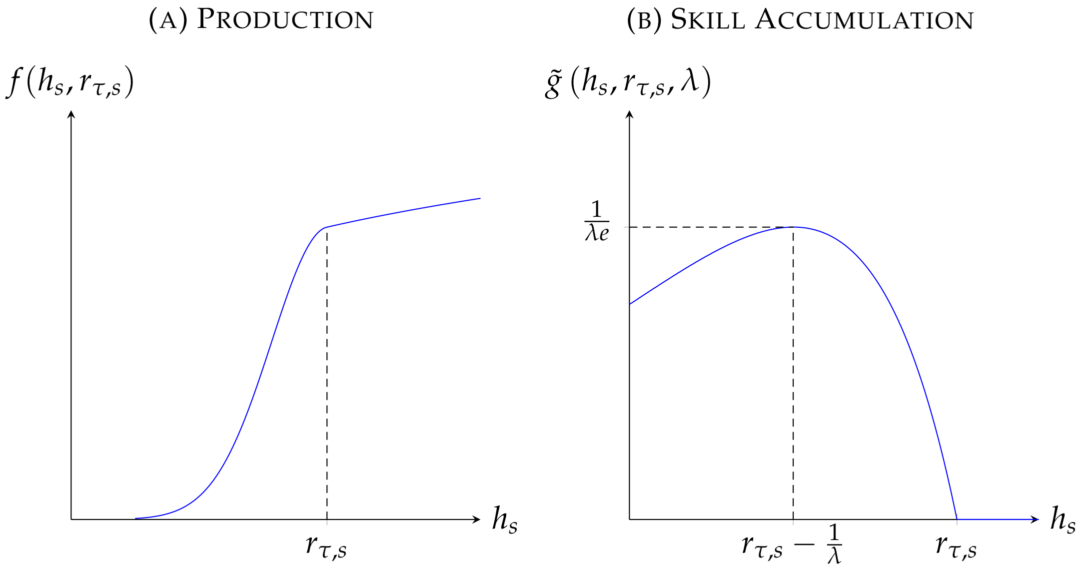
Appendix

FIGURE A.1: PRODUCTION AND SKILL ACCUMULATION: FUNCTIONAL FORMS

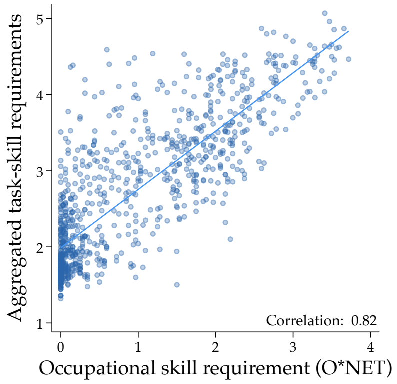


Notes: This figure illustrates the functional forms of the production and skill accumulation functions in equations (14) and (16), respectively. Panel A shows the production function $f(h_s, r_{\tau,s}) = h_s^{\omega} \exp\left(-\eta \min\{h_s - r_{\tau,s}, 0\}^2\right)$. Panel B shows the learning part of the skill accumulation function: $\tilde{g}(h_s, r_{\tau,s}, \lambda) = \max\{r_{\tau,s} - h_s, 0\} \exp(-\lambda \max\{r_{\tau,s} - h_s, 0\})$. It illustrates that maximum learning is attained when the skills are $\frac{1}{\lambda}$ below the skill requirements.

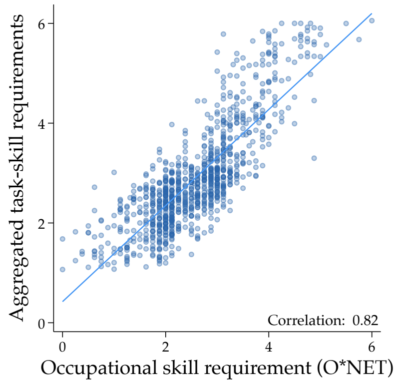
A Figures

FIGURE A.2: VALIDATION OF TASK SKILL REQUIREMENT DATA WITH O*NET

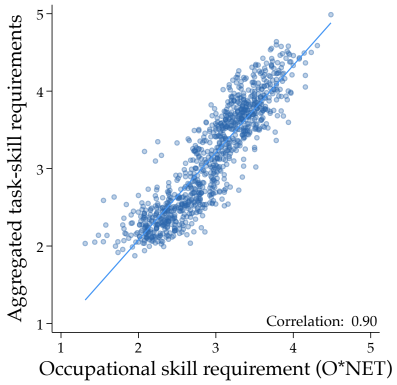
(A) MANUAL



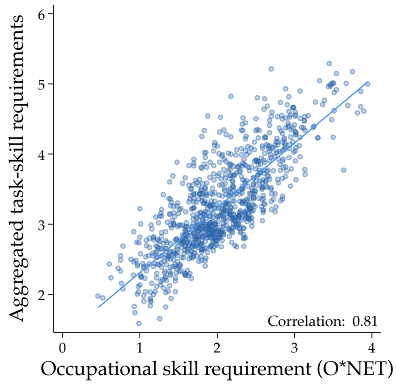
(B) MATH



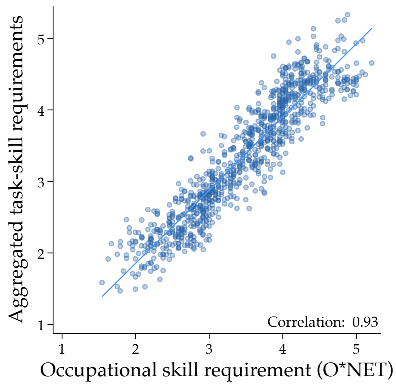
(C) SOCIAL



(D) TECHNICAL

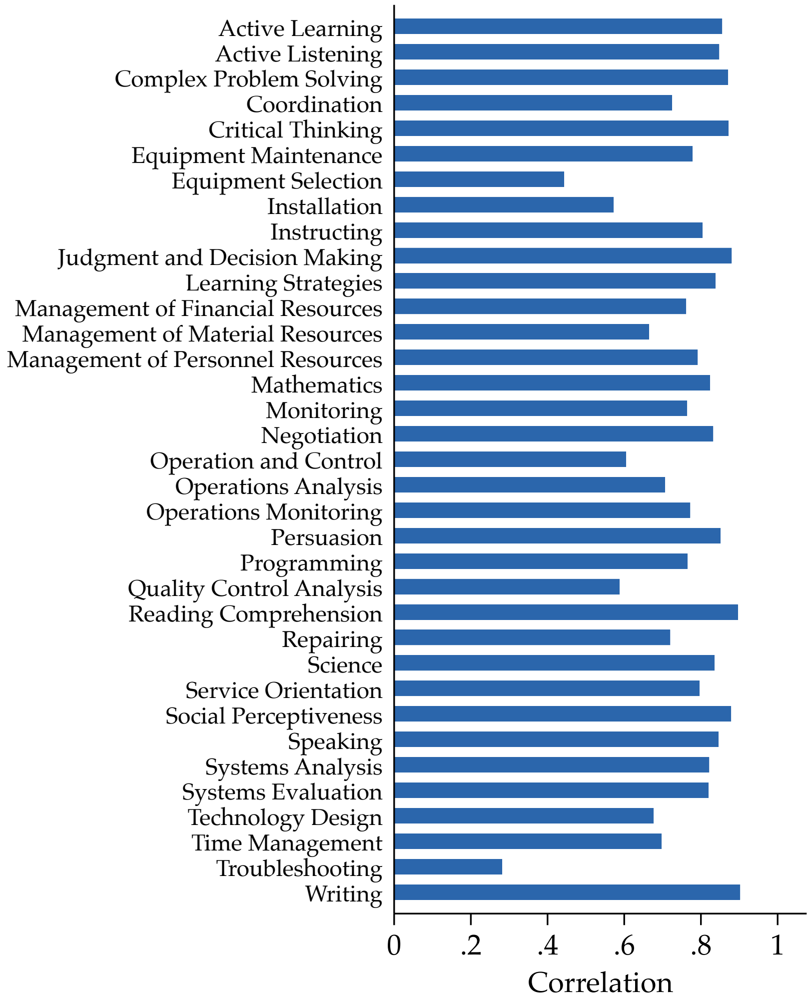


(E) VERBAL



Notes: This figure shows the correlation between the occupation-level skill requirement in the O*NET database and the GPT-4o generated task-level skill requirements aggregated to the occupation-level for the skills used in the analysis. Each observation represents an occupation in the O*NET database.

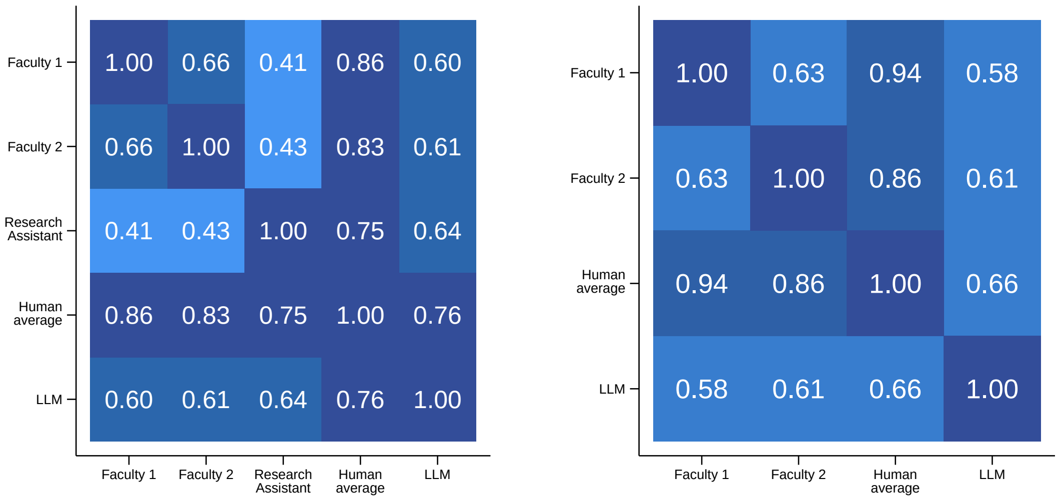
FIGURE A.3: VALIDATION OF TASK SKILL REQUIREMENT DATA WITH O*NET (35 SKILL DIMENSIONS)



Notes: This figure shows the correlations between the occupation-level skill requirement in the O*NET database and the GPT-4o generated task-level skill requirements aggregated to the occupation-level for each of the 35 skills.

Figure A.4: Human Validation of Task-Level Skill Requirements

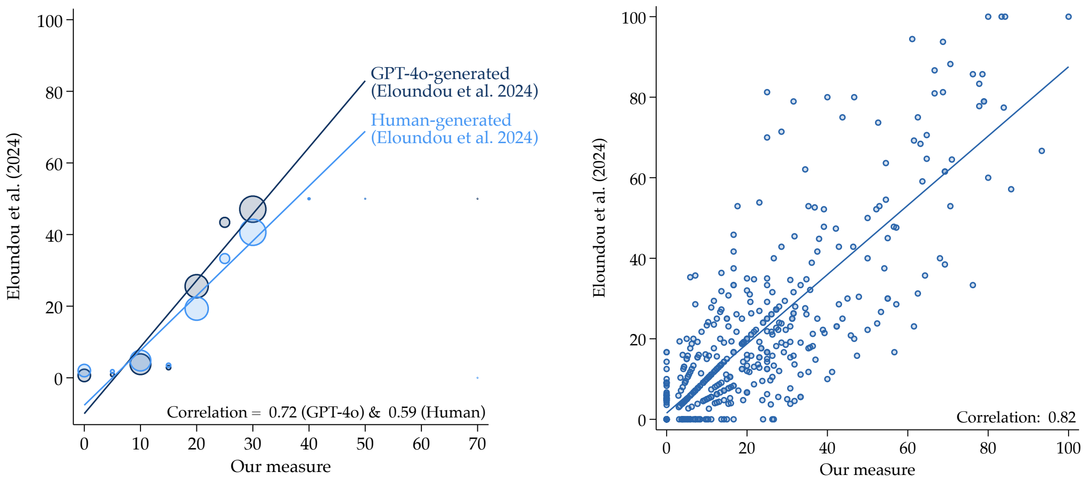
(a) Without AI — (b) With AI



Notes: This figure shows pairwise correlations of task-level skill requirement ratings across human raters and our LLM-generated ratings. Panel A shows pre-AI skill requirement levels for tasks in economics across three human raters along with the human average and the LLM. Panel B shows post-AI skill requirement levels for two human raters, the human average, and the LLM. All ratings are on a 1–7 scale used by O*NET.

Figure A.5: Agreement on AI exposure with Eloundou et al. (2024)

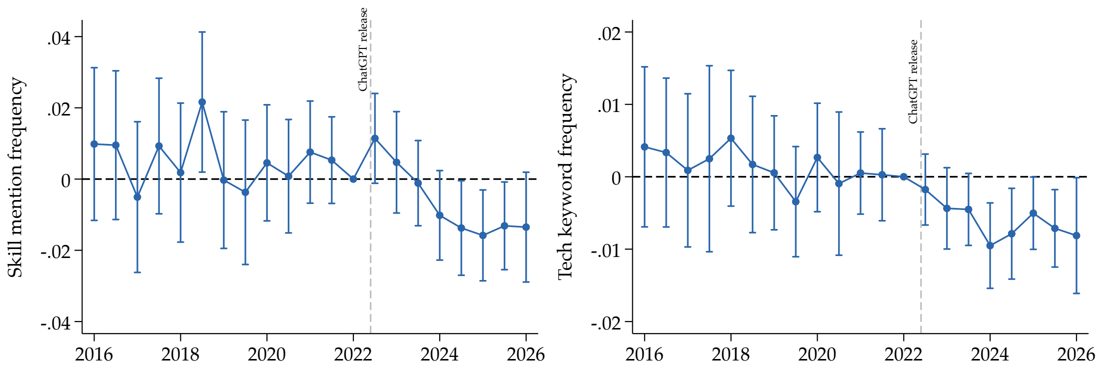
(a) Augmentation — (b) Automation



Notes: This figure compares our estimates tasks' exposure to augmentation and automation by generative AI. Panel A shows our task-level augmentation estimates (time saved to complete each task in an occupation) to those provided by Eloundou et al. (2024), measuring whether or not large language models can save at least 50 percent of time to complete a task (binary). Panel B shows the the share of tasks in each occupation that can be automated by generative AI with similar data provided by Eloundou et al. (2024).

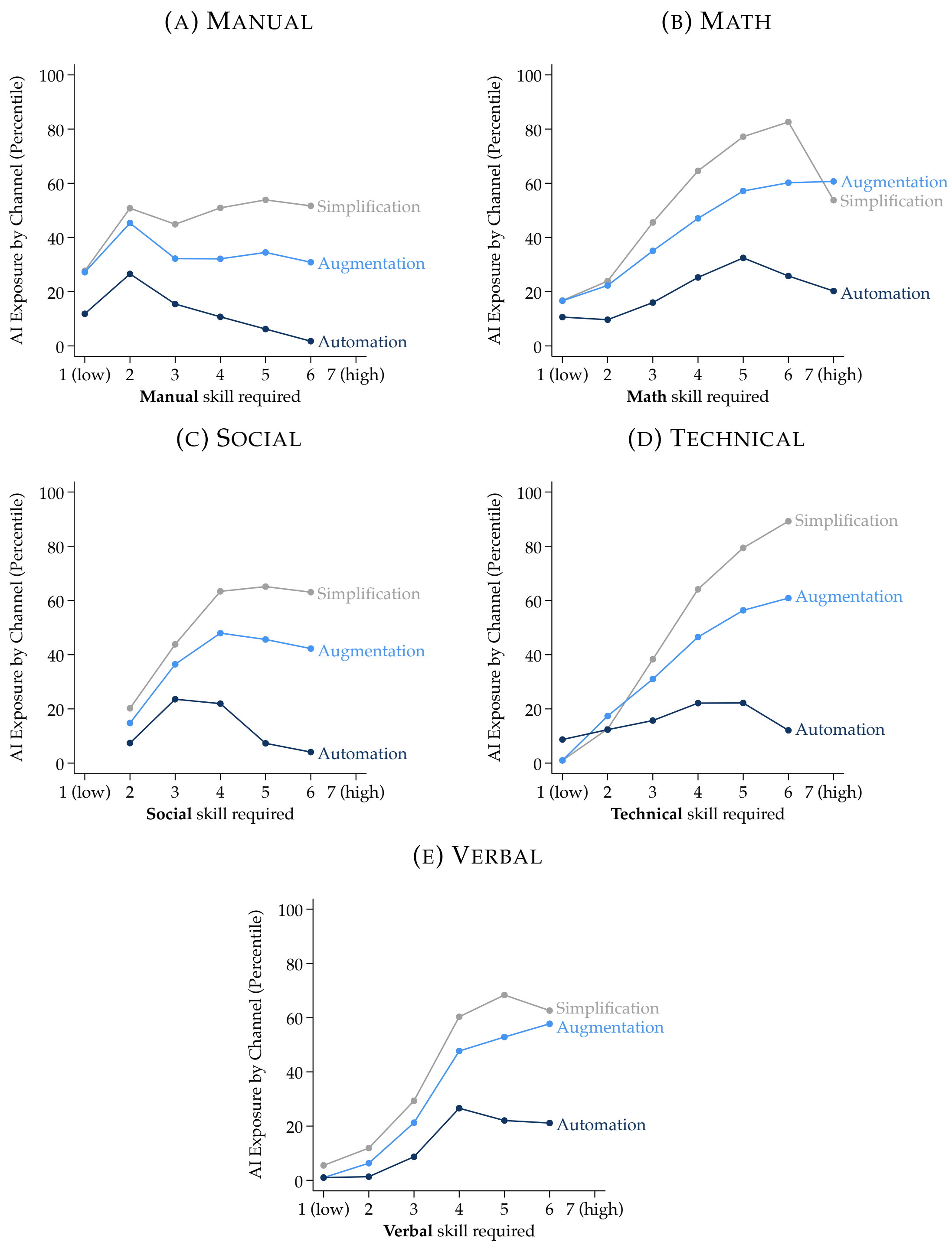
FIGURE A.6: ADDITIONAL MEASURES OF SIMPLIFICATION VIA JOB POSTINGS

(A) SKILL MENTION RATE (B) TECH KEYWORD RATE



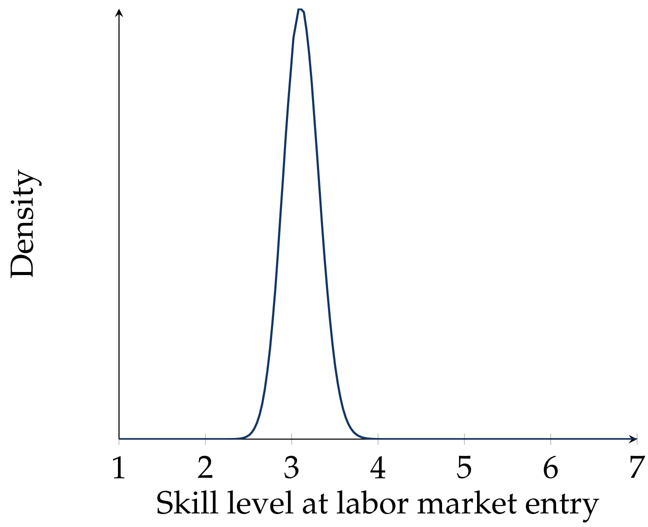
Notes: This figure shows that occupations predicted to be more simplified by AI show changes in the use of skill and technical language in real-time job postings. Continuous treatment exposure is measured using our occupation-level predicted AI-led simplification. Panel A uses the rate of the words "skill" or "skills" per 100 words, and Panel B uses the rate of technical keywords per 100 words (incl. terms like Python, Git, JavaScript, Excel, deep learning, LLM, or API). We use a random subsample of US job postings in Revelio Labs' Cosmos database. Each coefficient gives the differential effect of a 1 standard deviation higher predicted simplification in t relative to 2022-H1. We include occupation and time fixed effects; standard errors are clustered at the occupation level.

FIGURE A.7: SKILLS & GENERATIVE AI EXPOSURE BY CHANNEL



Notes: This figure shows the correlation between a task's skill requirements and its potential to be augmented, automated, or simplified by Generative AI. Each dot represents the average percentile of exposure to each channel among tasks with the same requirement in a given skill.

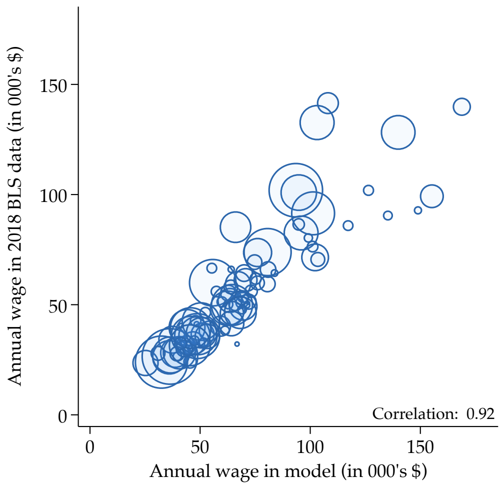
Figure A.8: Initial skill distribution



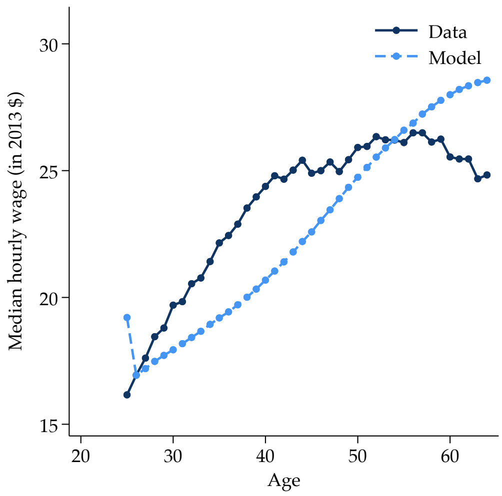
Notes: This figure shows the estimated density of the skill distribution of young workers (age $a = 1$ in the model) on O*NET's 1 to 7 scale. For comparison, for the skill "reading comprehension", a 2 means being able to "read step-by-step instructions for completing a form", 4 means being able to "understand an email from management describing new personnel policies", and 6 means being able to "read a scientific journal article describing surgical procedures".

Figure A.9: Model Fit: Comparing Model Moments with Data

(A) Occupational Wages

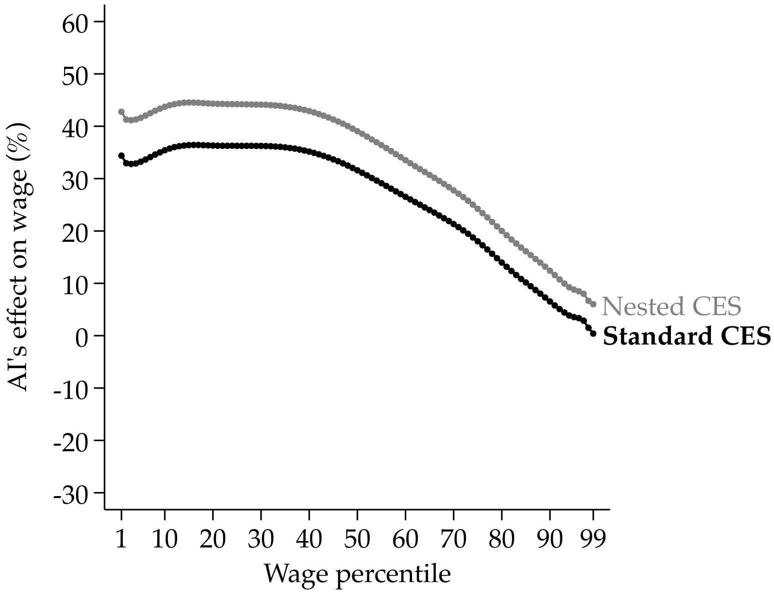


(B) Life-cycle of wages



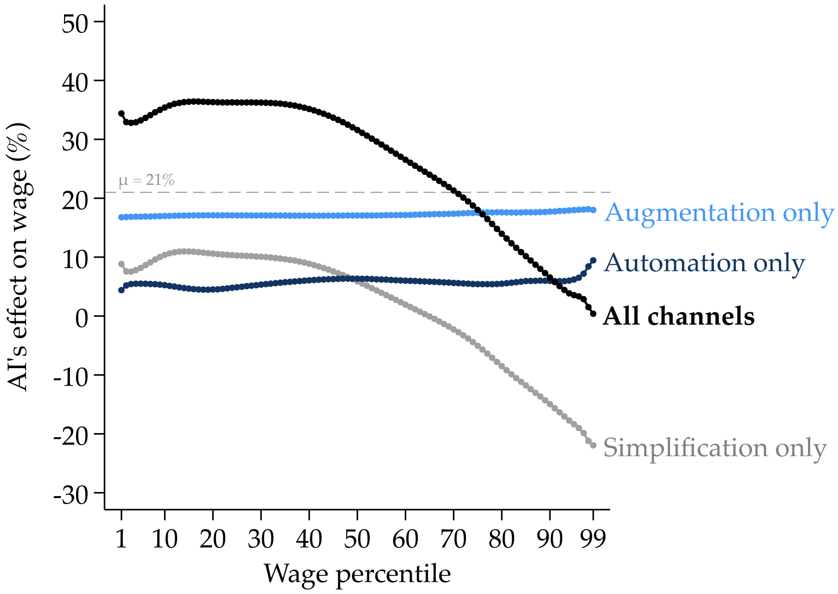
Notes: Panel A shows the correlation between the average wage in an occupation in the model's steady state and in the data as reported in the 2018 BLS OEWS data. Panel B reports the median wage by age in the NLSY79 and the model's steady state.

FIGURE A.10: WAGE EFFECTS: STANDARD VS. NESTED CES



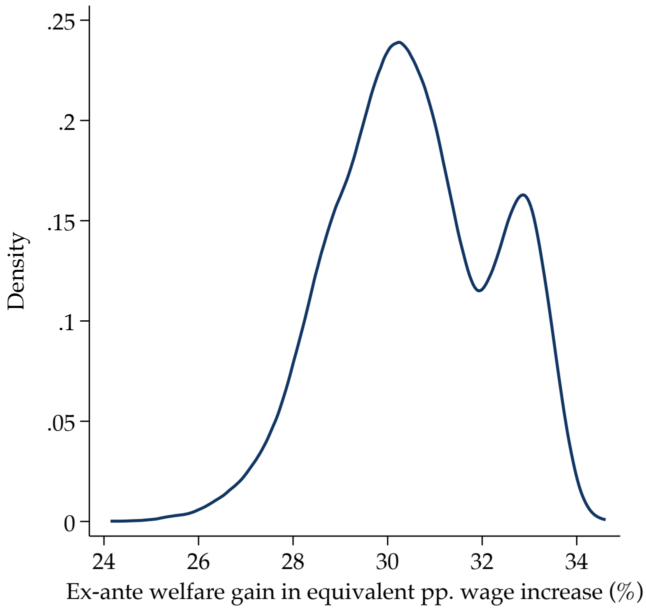
Notes: This figure shows the distribution of wage changes induced by generative AI across the wage percentile distribution under two alternative demand specifications. The horizontal axis represents wage percentiles weighted by pre-AI employment, and the vertical axis shows the percentage change in wages for each percentile. The black line shows results under the baseline one-level CES demand structure, while the gray line shows results under a nested CES structure with three layers: broad industry groups, detailed industries, and occupations. Both specifications include the joint effect of AI's augmentation, automation, and simplification.

FIGURE A.11: WAGE EFFECTS BY CHANNEL



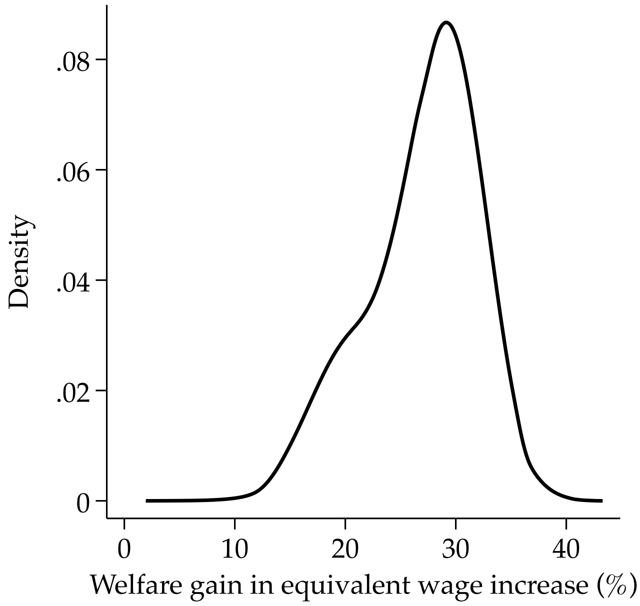
Notes: This figure shows the distribution of wage changes induced by generative AI across the wage percentile distribution. The horizontal axis represents wage percentiles weighted by pre-AI employment, and the vertical axis shows the percentage change in wages for each percentile. The black line shows the joint effect of AI's augmentation, automation, and simplification on each wage percentile. Other lines show the effects when each channel operates alone.

FIGURE A.12: DISTRIBUTION OF LONG-RUN WELFARE EFFECTS



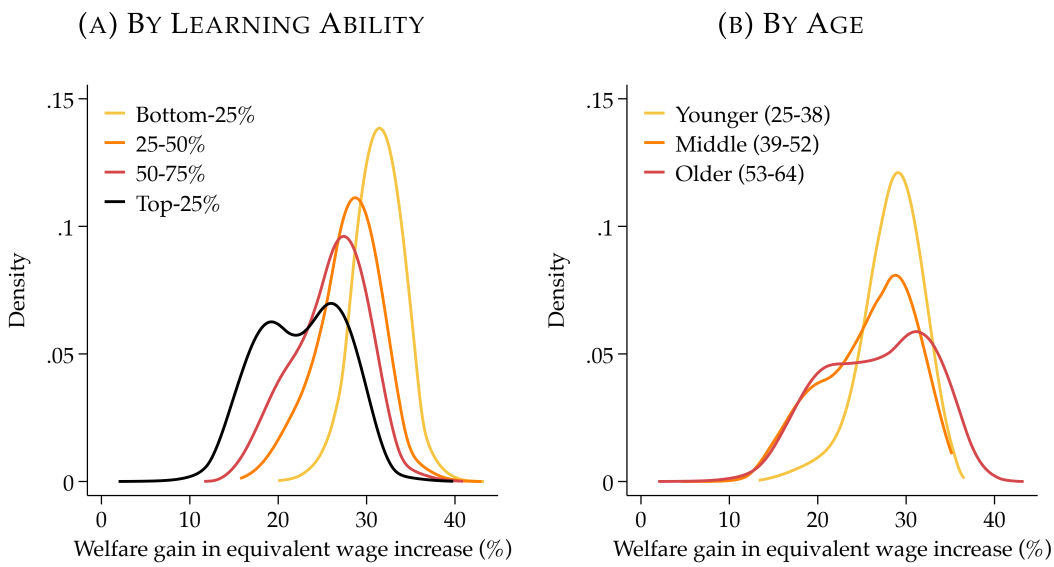
Notes: This figure shows the distribution of AI's welfare effect on individual workers. The welfare effect is measured in equivalent wage variation: it represents the permanent wage increase across all occupations that yields the same welfare gain as the introduction of AI.

FIGURE A.13: DISTRIBUTION OF SHORT-RUN WELFARE EFFECTS



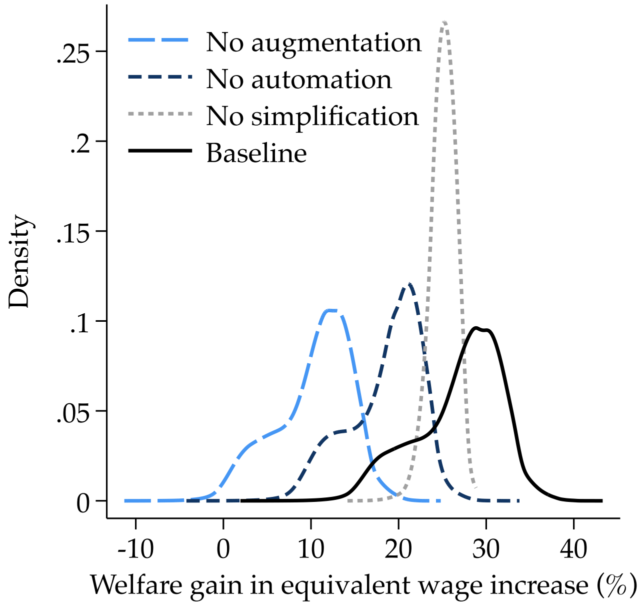
Notes: This figure shows the distribution of AI's welfare effect on individual workers along the transition path (at the time of the shock). The welfare effect is measured in equivalent wage variation: it represents the permanent proportional wage increase across all occupations that would yield the same welfare gain as the introduction of AI. Unlike the steady-state comparison, this measure accounts for the dynamic adjustment path of the economy.

FIGURE A.14: SHORT-RUN WELFARE EFFECTS BY WORKER CHARACTERISTICS



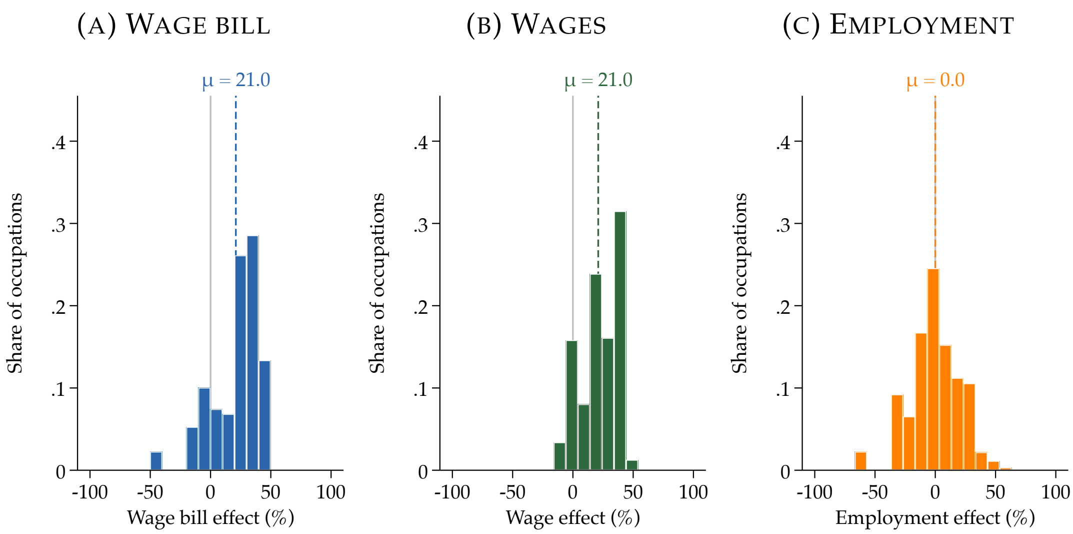
Notes: This figure shows heterogeneity in AI's welfare effects along the transition path by worker characteristics. Panel (A) shows welfare effects by learning ability (AFQT score). Panel (A) shows welfare effects by age. The welfare effect is measured in equivalent wage variation: it represents the permanent wage increase across all occupations that yields the same welfare gain as the introduction of AI.

FIGURE A.15: DISTRIBUTION OF SHORT-RUN WELFARE EFFECTS UNDER DIFFERENT SCENARIOS



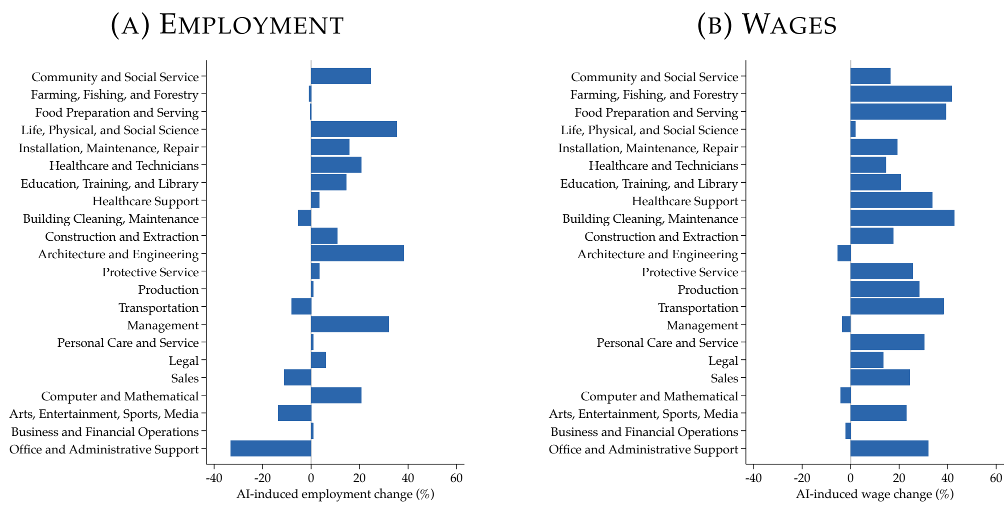
Notes: This figure shows the distribution of AI's welfare effect on individual workers along the transition path under different scenarios. The dark line represents the distribution under the baseline scenarios. The other lines show the distribution when one of the channels is not present. The welfare effect is measured in equivalent wage variation: it represents the permanent wage increase across all occupations that yields the same welfare gain as the introduction of AI. Unlike the steady-state comparison, this measure accounts for the dynamic adjustment path of the economy.

Figure A.16: Generative AI's effect across occupations



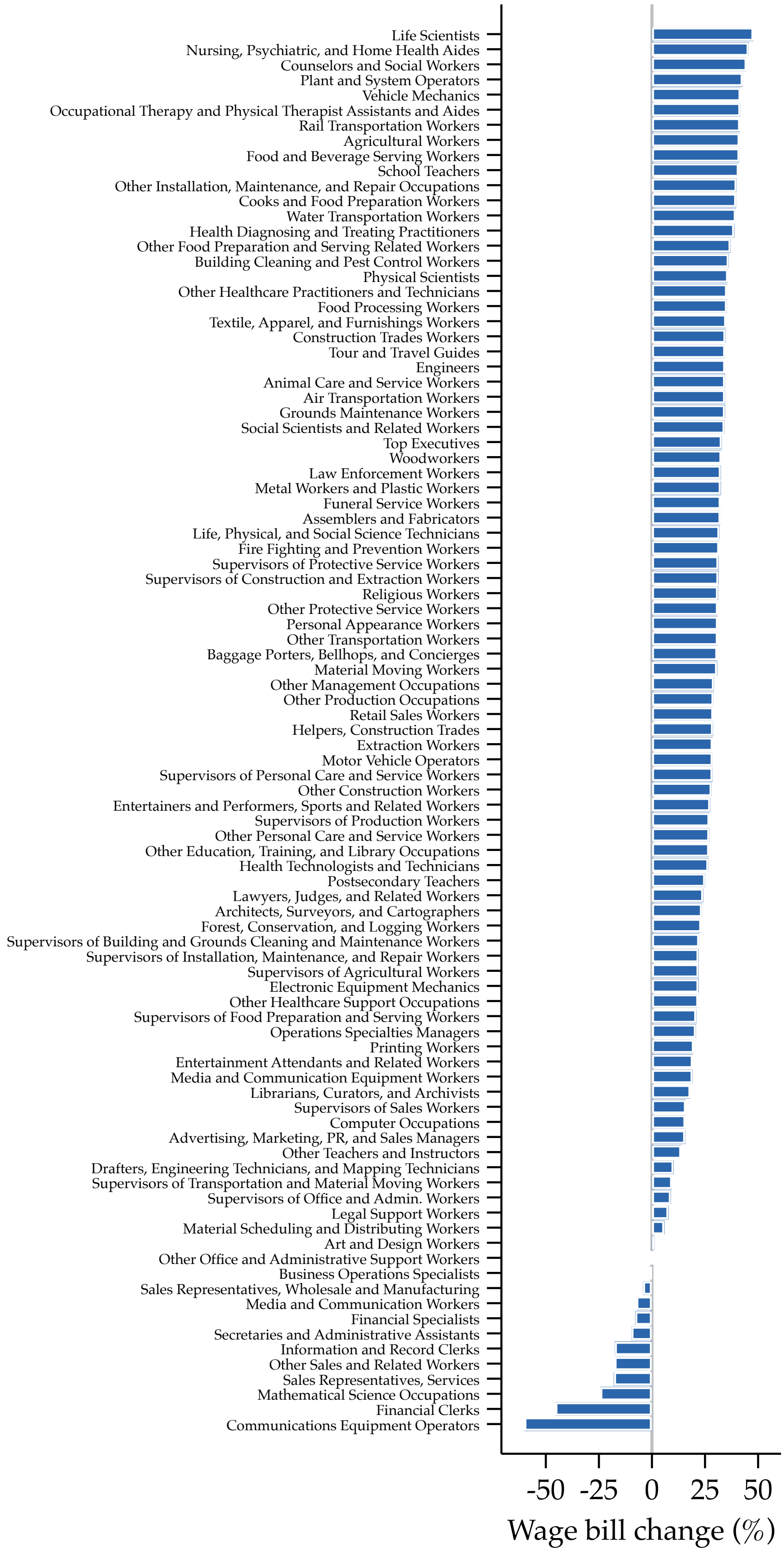
Notes: This figure shows the distribution of generative AI's predicted effects across occupations based on our structural model. Panel (A) shows wage bill changes (wages × employment). Panel (B) shows wage changes. Panel (C) shows employment effects, which are symmetric around zero by definition as our model does not feature unemployment.

Figure A.17: AI's effect on occupational employment & wages



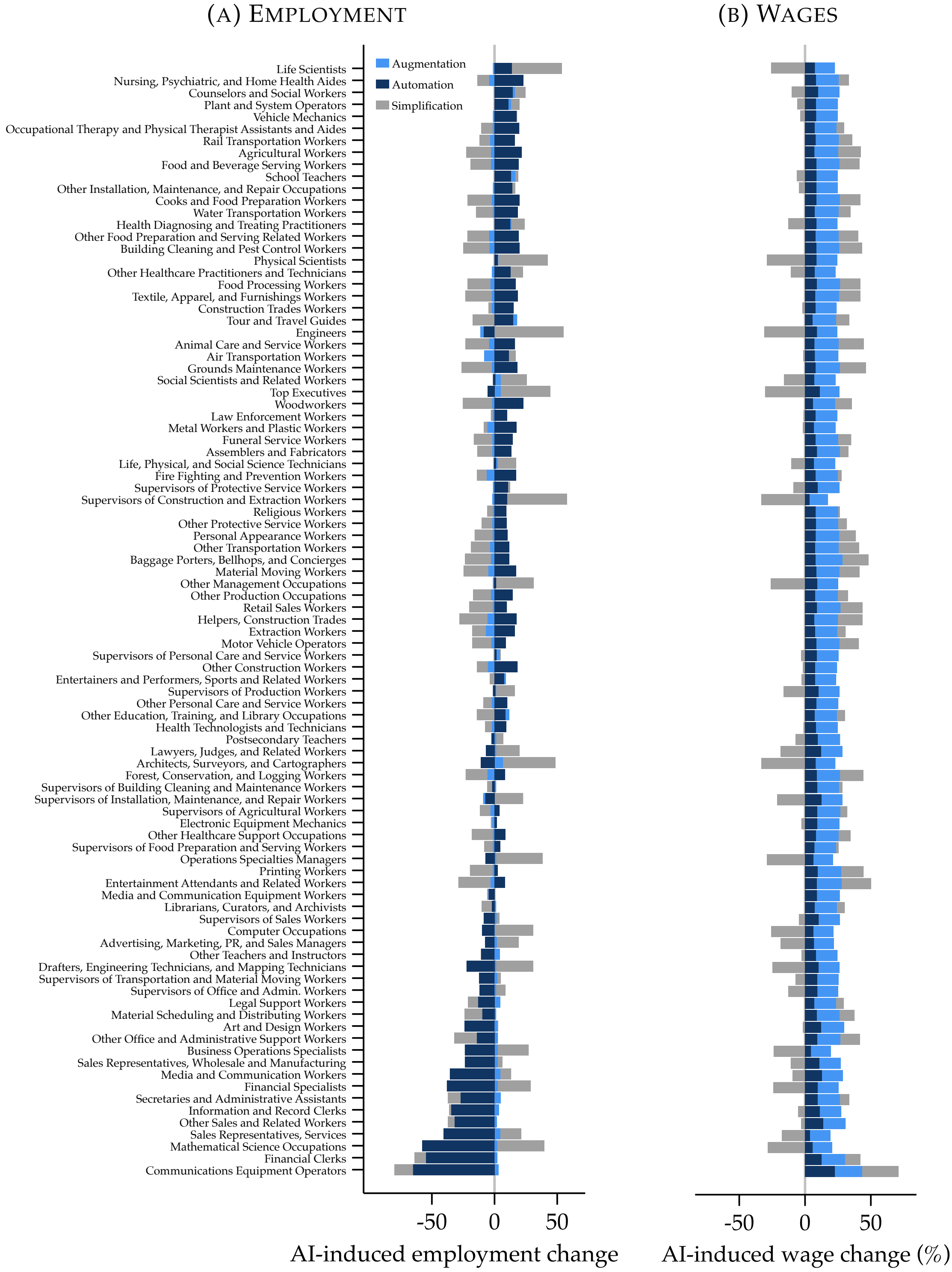
Notes: This figure shows the model's predictions on AI's employment and wage effects by occupational group. Occupations are sorted in descending order of AI's effect on their wage bill, so that the first listed occupation experiences the largest wage bill increase.

FIGURE A.18: AI'S EFFECT ON DETAILED OCCUPATIONS' WAGE BILLS



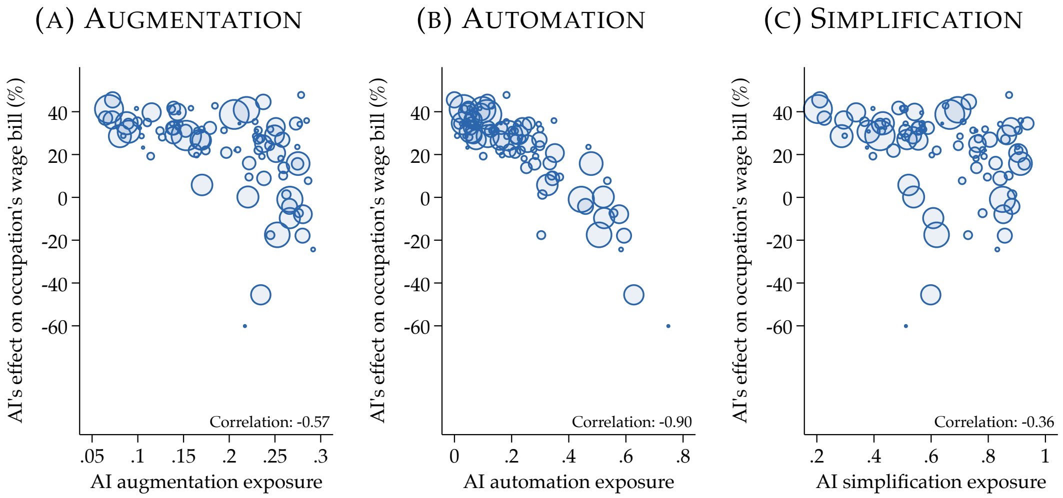
Notes: This figure shows the model predictions on AI's wage bill effects by occupation.

FIGURE A.19: AI'S EFFECT ON DETAILED OCCUPATIONS' EMPLOYMENT & WAGES



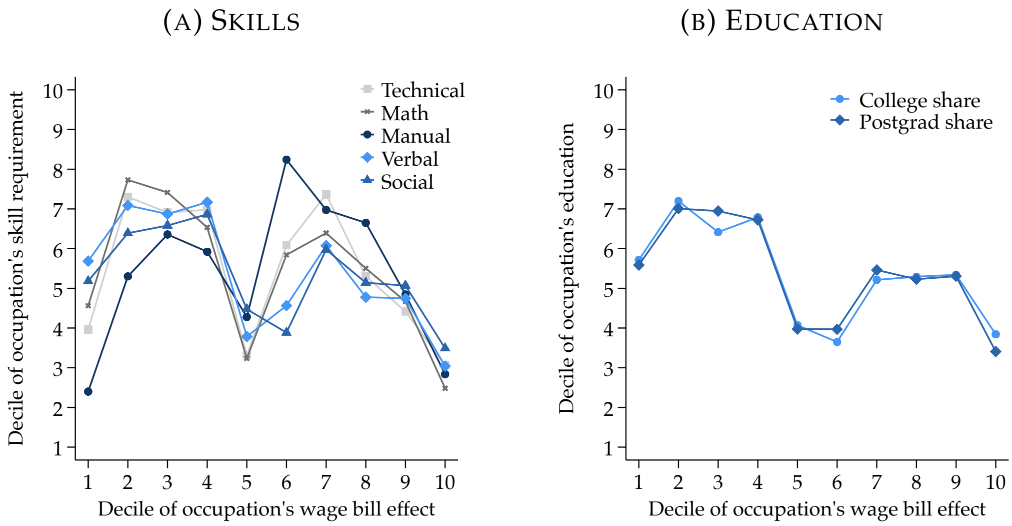
Notes: This figure shows the model's predictions on AI's employment and wage effects by occupation. Occupations are sorted in descending order of AI's effect on the total wage bill, so that the first listed occupation experiences the largest wage bill increase. We conduct a Shapley-Owen decomposition to separate the overall change into the contribution of each channel: augmentation, automation, and simplification.

FIGURE A.20: AUTOMATION EXPOSURE MOST PREDICTIVE OF LABOR MARKET LOSSES



Notes: This figure shows the relationship between three dimensions of AI exposure and model-predicted changes in occupational wage bills. Each bubble represents an occupation, with size proportional to pre-AI employment. Panel (A) shows augmentation exposure, measured as the share of time access to generative AI can save in completing an occupation's tasks. Panel (B) shows automation exposure, measured as the share of an occupation's tasks than generative AI can complete autonomously. Panel (C) shows simplification exposure, measured as the (negative) relative change of skill levels required to complete an occupation's tasks (averaged across all 35 O*NET skills).

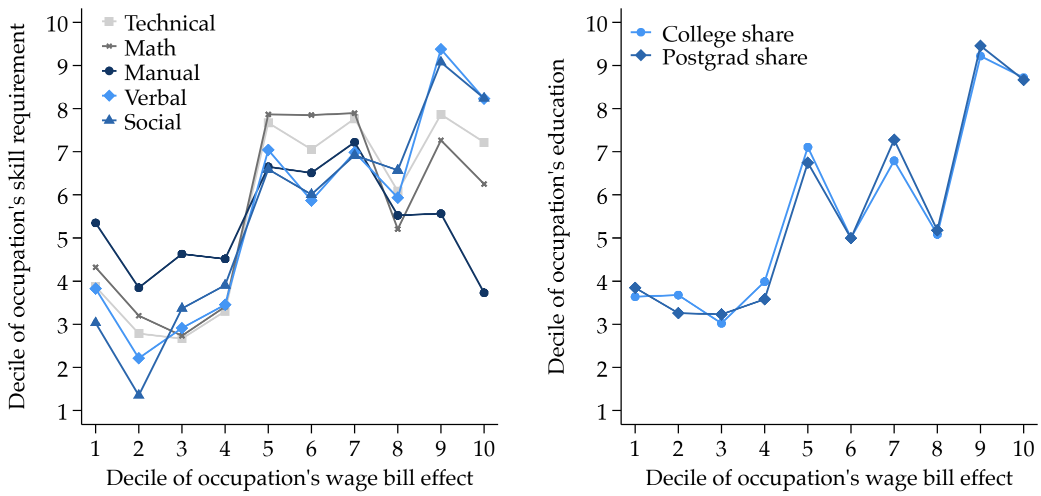
FIGURE A.21: SKILLS AND EDUCATION BY GENAI'S WAGE BILL EFFECT



Notes: This figure shows the relationship between occupational characteristics and generative AI's wage bill effects. Panel A plots average skill requirement deciles against wage bill effect deciles. Panel B plots education levels against wage bill effect deciles. Each point represents a decile of occupations ranked by their predicted wage bill change, weighted by pre-AI employment.

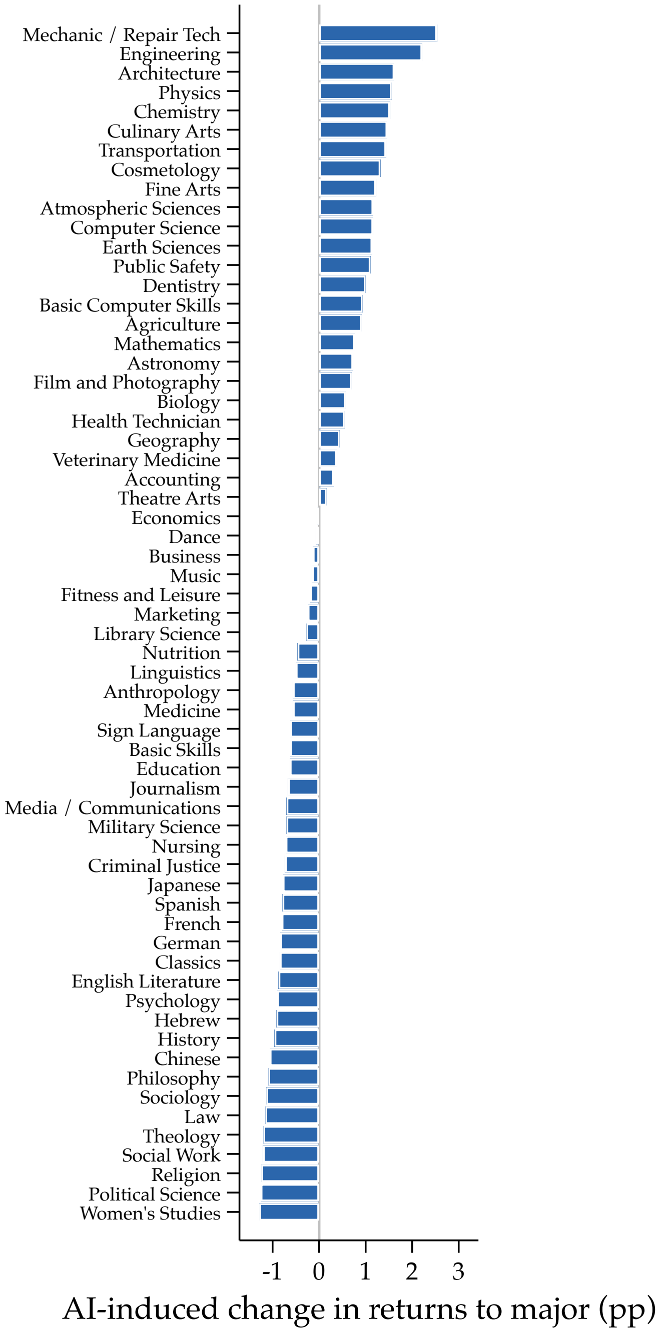
FIGURE A.22: SKILLS AND EDUCATION BY WAGE BILL EFFECT: PHYSICAL AI SCENARIO

(A) SKILLS (B) EDUCATION



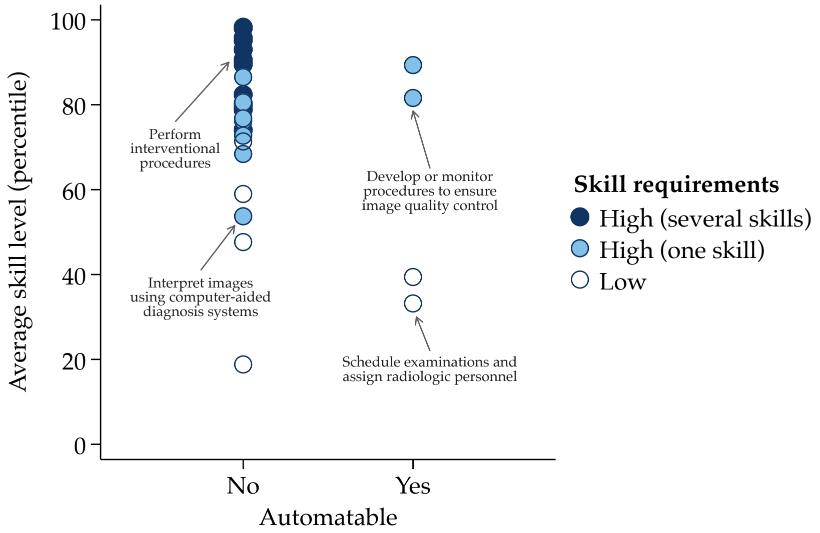
Notes: This figure shows the relationship between occupational characteristics and AI's wage bill effects when physical capabilities are available. Panel A plots average skill requirement deciles against wage bill effect deciles. Panel B plots education levels against wage bill effect deciles. Each point represents a decile of occupations ranked by their predicted wage bill change, weighted by pre-AI employment.

FIGURE A.23: AI'S EFFECT ON RETURNS TO COLLEGE MAJORS



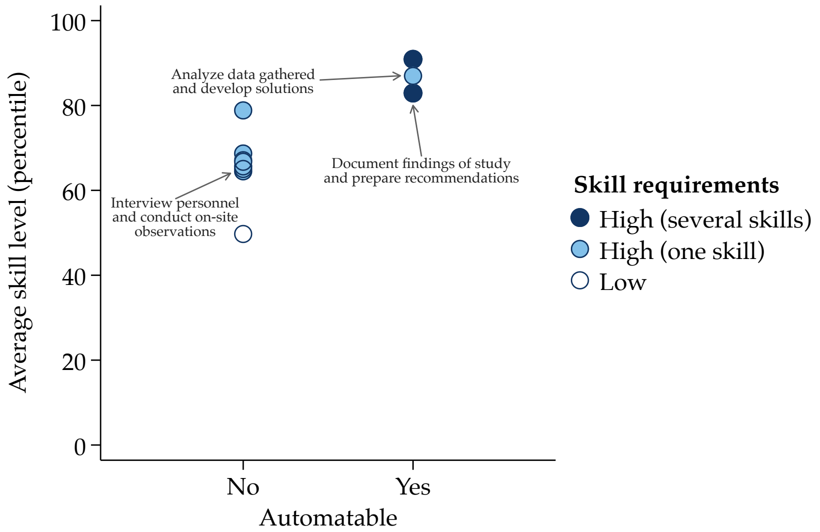
Notes: This figure shows the model's predictions on AI's effect on returns to college majors. Returns are calculated by combining major-level skill intensities from the Skill Atlas with the model's predicted changes in returns to each skill dimension. Values are centered relative to the average major.

Figure A.24: Radiologists' Task-Based Automation Exposure



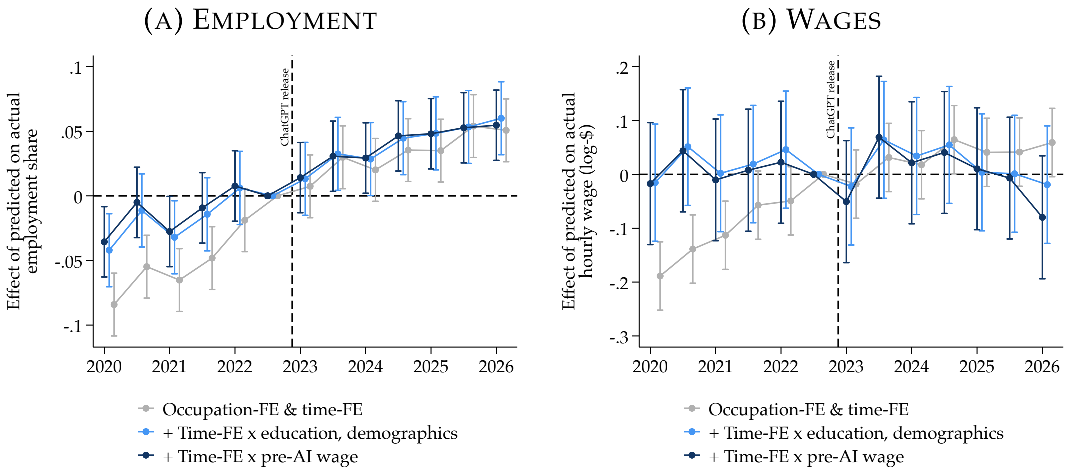
Notes: This figure shows radiologists' automation exposure across all tasks they engage in according to O*NET. We classify a task as 'automatable' if it has high or full automation exposure, which is restricted to cases where the LLM indicates that generative AI can complete at least 90% of the components of the tasks. We classify skill requirements as "high" if they exceed level 3.5 in O*NET's scale from 1 to 7.

Figure A.25: Management Analysts' Task-Based Automation Exposure



Notes: This figure shows management analysts' automation exposure across all tasks they engage in according to O*NET. We classify a task as 'automatable' if it has high or full automation exposure, which is restricted to cases where the LLM indicates that generative AI can complete at least 90% of the components of the tasks. We classify skill requirements as "high" if they exceed level 3.5 in O*NET's scale from 1 to 7.

FIGURE A.26: EARLY LABOR MARKET EFFECTS OF GENERATIVE AI



Notes: Event study estimates ($\widehat{\beta}_k$) show differential changes in occupational employment and wages following ChatGPT's November 2022 release. Coefficients represent the effect of the model-predicted change in the outcome on the observed outcome. A coefficient of 1.0 would indicate complete realization of model predictions. Estimates use CPS data aggregated to 6-month periods with occupation fixed effects and time fixed effects. The specification with controls includes occupation-specific trends based on education, sectoral composition, demographics, and pre-AI wages. Error bars represent 95%-confidence intervals.

B Tables

TABLE B.1: AGGREGATION OF O*NET'S SKILL REQUIREMENTS TO 5 DIMENSIONS

Skill	O*NET skill	O*NET skill category
Manual	Equipment Maintenance	Technical
	Equipment Selection	Technical
	Installation	Technical
	Repairing	Technical
Math	Mathematics	Basic Content
Social	Active Listening	Basic Content
	Coordination	Social
	Instructing	Social
	Management of Personnel Resources	Resource Management
	Negotiation	Social
	Persuasion	Social
	Service Orientation	Social
	Social Perceptiveness	Social
Technical	Complex Problem Solving	Complex Problem Solving
	Judgment and Decision Making	Systems
	Operation and Control	Technical
	Operations Analysis	Technical
	Operations Monitoring	Technical
	Programming	Technical
	Quality Control Analysis	Technical
	Science	Content
	Systems Analysis	Systems
	Systems Evaluation	Systems
	Technology Design	Technical
	Troubleshooting	Technical
Verbal	Reading Comprehension	Basic Content
	Speaking	Basic Content
	Writing	Basic Content

Notes: This table shows the mapping of the five skill clusters—Manual, Math, Social, Verbal, and Technical—to the relevant O*NET skills and their respective O*NET's skill category. For each of the skills, we set the requirement to the average across the relevant O*NET skills. We dropped 7 out of 35 O*NET skill dimensions that could not be clearly mapped into the skills used in the analysis.

TABLE B.2: AGREEMENT ON AUTOMATION EXPOSURE WITH ELOUNDOU ET AL. (2024)

Our measure	**Eloundou et al. (2024)** None	Low	Medium	High	Full
None	26.86	1.65	0.01	0.34	0.14
Low	6.12	21.34	1.98	2.18	0.12
Medium	0.17	7.27	7.61	3.65	0.06
High	0.04	1.77	5.61	11.53	0.18
Full	0.02	0.07	0.02	1.02	0.24

Notes: This table shows the agreement rates between our measure of automation and that of Eloundou et al. (2024) on the task-level. The table is computed based on 14,209 tasks (out of 19,530) that are in both databases. We classify a task as automated if the exposure is "high" or "full". The share of automated tasks is 21% and 22% in Eloundou et al. (2024)'s and our measure, respectively.

TABLE B.3: SUMMARY OF TASK-LEVEL DATA ON AI CAPABILITIES

	Augmentation		Automation	Simplification
	Excluding automatable tasks	Including automatable tasks		
Mean	17.9%	20.2%	22.2%	18.3%
Std. Dev.	9.4%	9.6%	41.6%	6.6%
Median	20.0%	20.0%	0.0%	20.4%
Range	0.0% - 70.0%	0.0% - 70.0%	0.0% - 100.0%	0.0% - 32.0%
Tasks	15,192	19,530	19,530	19,530

Notes: This table summarizes our new estimates of generative AI's potential impact on tasks across three channels: augmentation (share of worker's time saved by technology to complete the task), automation (share of tasks that can be fully automated by technology), and simplification (relative decrease in average skill requirements across all 35 O*NET skill dimensions). For augmentation, we present estimates both excluding and including tasks that can be automated. Augmentation and simplification estimates are generated by GPT-4o; automation estimates are generated by GPT-5 with low to medium reasoning effort.

TABLE B.4: EXPERIMENTAL ESTIMATES COMPARED TO OUR TASK AUGMENTATION DATA

Occupation	Task	Tool	Estimate		N	Notes	Source
			Theirs	Ours			
Software developer	Coding	GitHub Copilot	26%	30%	4,867	+26.1% number of completed tasks in lab; new developers higher adoption rates & higher productivity gains	[1]
Software developer	Coding	GitHub Copilot	56%	27%	95	55.8% time saved, quality ↑	[2]
Software developer	Coding	GitHub Copilot	36%	30%	23	36% time saved for familiar tasks; no change for unfamiliar tasks; 48% fewer issues	[3]
Programmer	Coding	GPT-3	27%	30%	100	27% time saved among 100 expert programmers; 50 non-programmers perform tasks similarly well with LLM	[4]
Programmer	Coding	GitHub Copilot	0%	28%	24	No time saved; however, most participants still preferred using LLM	[5]
Management consultant	Consulting	GPT-4	25%	30%	758	25.1% time saved, +12.2% tasks completed, +40% quality (decreased for tasks beyond AI frontier); lower-skilled consultants benefited more	[6]
Customer support	Resolution	GPT-4	14%	30%	5,179	+14% productivity (issues resolved per hour), +34% for new & low-skill workers; minimal impact on experienced & high-skill workers	[7]
–	Writing	GPT-3.5	40%	30%	453	40% time saved, +18% output quality; inequality between workers ↓; low-skill workers benefited most; likelihood of using AI after experiment ↑	[8]
Taxi driver	Selecting routes	AI Navi	14%	9%	520	Shorter cruising time; gains only among low-skill drivers	[9]
Lawyer	Legal writing	Vincent & o1-preview	20%	30%	127	19.9% time saved across different legal writing tasks, quality ↑, LLM "Vincent" slightly higher gains	[10]
Product designer	Product marketing & development	GPT-4o	13-16%	30%	776	+0.37 SD quality and 16.4% time saved for individuals; +0.39 SD quality and 12.7% time saved for teams	[11]
Software developer	Coding	GitHub Copilot	65%	27%	24	Developers implemented ∼65% more requirements with AI assistance	[12]
Software developer	Coding	Google AI Tools	21%	30%	96	AI users finished an enterprise-grade task 21% faster. Results stronger for senior developers.	[13]
Programmer	Coding	CodeFuse	55%	30%	1,219	Lines of code produced ↑ 55%, gains concentrated among junior staff	[14]
Knowledge workers	E-mail	MS 365 Copilot	11%	26%	7,137	Treated spent 12% less time on email each week; did not significantly change time spent in meetings.	[15]
Train commissioning technician	Trouble-shooting	GPT-3.5 + RAG	20%	20%	173	+1.14 SD quality score; 20% increase in tasks completed not significant; less-experienced benefit more.	[16]
Knowledge workers	E-mail	GPT-4.1 assistant	9–15%	26%	1,174	Duration falls by 0.96 min (low edu) and 1.51 min (high edu) on ∼10.4–10.7 min baseline (∼9–15%); closes ∼75% of edu performance gap.	[17]

Notes: Sources correspond to [1] Cui et al. (2024), [2] Peng et al. (2024), [3] Clarke and Hanrahan (2024), [4] Campero et al. (2022), [5] Vaithilingam et al. (2022), [6] Dell'Acqua et al. (2023), [7] Brynjolfsson et al. (2025), [8] Noy and Zhang (2023), [9] Kanazawa et al. (2022), [10] Schwarcz et al. (2024), [11] Dell'Acqua et al. (2025), [12] Weber et al. (2024), [13] Paradis et al. (2024), [14] Gambacorta et al. (2024), [15] Dillon et al. (2025), [16] Lowhagen et al. (2025), [17] Cruces et al. (2026). To construct our own estimates of task augmentation (share of time saved to complete task) by generative AI at the level of work activities, we aggregate our task-level estimates within the relevant occupation as equally weighted averages for tasks we judge to be relevant to the work activity covered in each experiment.

TABLE B.5: OVERVIEW OF MODEL PARAMETERS

Model object	**Symbol**	**Value**	**How it is set**
Elast. of substitution: Occupations	σ	1.57	Burstein et al. (2019); Caunedo et al. (2023).
Elast. of substitution: Tasks	ρ	0.49	Humlum (2021).
Number of occupations	J	93	3-digit BLS SOC occupations.
Number of periods	A	40	Years between 25 and 65.
Discount factor	β	0.78	Following Keane and Wolpin (1997).
Skill dimensions	S		Addison et al. (2020); Baley et al. (2022), plus manual.
Occupational task sets	$\mathcal{T}_j$		O*NET tasks.
Occupational task weights	$\theta_{j,\tau}$		O*NET task importance.
Task-level skill requirements	$\boldsymbol{r}_\tau$		Large language model.
Task-level AI augmentation	γ_τ		"
Task-level AI automation	$\mathcal{A}_j$		"
Learning cost: 1st AFQT quartile	$\lambda(1)$	3.50	Maximum likelihood.
Learning cost: 2nd AFQT quartile	$\lambda(2)$	2.97	"
Learning cost: 3rd AFQT quartile	$\lambda(3)$	2.81	"
Learning cost: 4th AFQT quartile	$\lambda(4)$	2.67	"
Human capital depreciation	δ	0.0003	"
Scale of productivity shocks	ζ	0.053	"
Occupational switching cost	κ	0.340	"
Skill distribution (Beta)	(B_a, B_b)	(62,114)	"
Cost of underqualification	η	0.04	OLS within MLE routine.
Skill productivity: Manual	ω_{Mn}	0.33	"
Math	ω_{Mt}	0.79	"
Social	ω_S	0.55	"
Technical	ω_T	0.22	"
Verbal	ω_V	0.36	"
Occupational amenities	$\{\mu_j\}_{j=1}^J$		Match employment shares à la (Berry et al., 1995).
Occupational demand	$\{\alpha_j\}_{j=1}^J$		Using estimated prices and wage bills.

Notes: This table provides an overview of the parameters of the model, their mathematical symbols, the value at which they are set, and the procedure with which we arrived at the value.

TABLE B.6: OCCUPATIONAL SKILL REQUIREMENTS PREDICT OCCUPATIONAL PRICES

Skill requirements	Dependent variable: Log occupational price $\hat{p}_j$			
Manual	0.228	0.230	0.226	0.227
	(0.207)	(0.214)	(0.201)	(0.208)
Math	-0.070	-0.073	-0.077	-0.080
	(0.277)	(0.286)	(0.269)	(0.278)
Social	-0.083	-0.078	-0.077	-0.073
	(0.352)	(0.363)	(0.341)	(0.353)
Technical	1.342**	1.341**	1.353**	1.352**
	(0.599)	(0.619)	(0.581)	(0.601)
Verbal	0.554	0.554	0.550	0.550
	(0.376)	(0.388)	(0.364)	(0.377)
Sample occupations	All	50+	All	50+
Empirical Bayes applied	No	No	Yes	Yes
Observations	93	87	93	87
R^2	0.73	0.73	0.74	0.74

Notes: This table shows the coefficients and R^2 of a regression of the estimated occupational prices (in logs) on occupational skill requirements. Each observation represents one occupation and is weighted by the number of worker-year observations in that occupation. Columns where sample occupations indicates "50+" only include occupations with at least 50 observations. The third and fourth column show the results with fixed effects on which empirical Bayes regression has been applied. Occupational prices are estimated as the occupational fixed effects in regression equation (17). Occupational skill requirements refer to $\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} r_{\tau,s}$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE B.7: MODEL FIT: WAGE INEQUALITY IN DATA AND MODEL

		Ratios			Top shares		
	Gini	$\frac{p90}{p10}$	$\frac{p90}{p50}$	$\frac{p75}{p25}$	10%	5%	1%
Data	0.32	4.01	2.13	2.04	0.26	0.16	0.05
Model	0.24	2.96	1.83	1.84	0.20	0.11	0.02

Notes: This table reports measures of inequality in the unconditional wage distribution in the data (NLSY79) and in the model's steady state. The unit of observation is a worker-age pair in the data and in the model. We only included workers who remain in the NLSY79 and work until 2020 without interruptions over 18 months. Sample weights are applied in the NLSY79 data.

TABLE B.8: SKILL CHANNELS AND LABOR MARKET OUTCOMES

	Wage growth (%)		Employment growth (%)	
	(1)	(2)	(3)	(4)
Augmentation	1.48	-2.89	2.11	6.79**
	(4.79)	(4.18)	(3.18)	(2.91)
Automation	4.74**	19.04***	-23.86***	-39.03***
	(1.80)	(3.94)	(1.39)	(3.67)
Simplification	-24.68***	-15.19***	24.81***	15.25***
	(4.25)	(3.38)	(2.93)	(2.94)
Indirect simplification				
Augmentation-led		3.52*		-2.50**
		(1.78)		(1.24)
Automation-led		-15.88***		17.05***
		(4.34)		(4.22)
Observations	93	93	93	93

Notes: This table presents weighted OLS regressions of occupation-level wage and employment growth on AI exposure measures and skill requirement changes. The dependent variable is wage growth in columns (1)–(2) and employment growth in columns (3)–(4). All independent variables are standardized to have standard deviation 1. Columns (1) and (3) include only direct AI exposure measures (augmentation, automation, simplification). Columns (2) and (4) add indirect skill requirement changes induced by AI. All regressions are weighted by pre-AI occupation employment shares. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Estimation

C.1 Cost savings and AI's income share

The cost of performing a task τ with the worker's unit of time equals

$$c_\tau^l(\boldsymbol{h}) \equiv \frac{\Lambda_j(\boldsymbol{h})}{\gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau)}$$

where $\Lambda_j(h)$ is the shadow value of a unit of time in occupation j given skills $\boldsymbol{h}$. Similarly, let c_τ^k be the unit cost of producing task τ with capital, i.e.,

$$c_\tau^k \equiv \frac{R}{\phi_\tau}.$$

For any automated task $\tau \in \mathcal{A}_j$, the cost of producing the task with capital relative to performing the task by labor thus equals

$$\chi_\tau \equiv \frac{c_\tau^k}{c_\tau^l(\boldsymbol{h})} = c_\tau^k \cdot \frac{\gamma_\tau f(\boldsymbol{h}, \boldsymbol{r}_\tau)}{\Lambda_j(\boldsymbol{h})}.$$

Since the shadow value of a unit of time is the wage, i.e., $\Lambda_j(\boldsymbol{h}) = w_j(\boldsymbol{h})$, equation (8) implies that cost savings are equal to

$$\chi_\tau = \frac{c_\tau^k}{p_j} \left(1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{c_\tau^k}{p_j}\right)^{1-\rho}\right)^{\frac{1}{\rho-1}} \frac{\gamma_\tau f\left(\boldsymbol{h}, \boldsymbol{r}_\tau\right)}{\left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} f(\boldsymbol{h}, \boldsymbol{r}_\tau)^{\rho-1}\right)^{\frac{1}{\rho-1}}}.$$

For our quantification, we need an estimate of $\sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{c_\tau^k}{p_j}\right)^{1-\rho}$ for each occupation. To obtain this, we make two simplifying assumptions. First, we assume that the cost savings do not vary across automatable tasks, i.e., $\chi_\tau = \chi$ for all $\tau \in \mathcal{A}_j$ and $\forall j = 1, \ldots, J$. Second, we assume that the automated tasks are not different in productivity and skill requirements from the non-automatable tasks, i.e., $\gamma_\tau^{\rho-1} f\left(\boldsymbol{h}, \boldsymbol{r}_\tau\right)^{\rho-1} \approx \sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} f(\boldsymbol{h}, \boldsymbol{r}_\tau)^{\rho-1}$ for all $\tau \in \mathcal{A}_j$ and $\forall j = 1, \ldots, J$. Under those two assumptions, the cost savings simplify to

$$\chi = \frac{c_\tau^k}{p_j} \left(1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{c_\tau^k}{p_j}\right)^{1-\rho}\right)^{\frac{1}{\rho-1}} \frac{1}{\left(1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau}\right)^{\frac{1}{\rho-1}}} \quad \forall j = 1, \ldots, J, \forall \tau \in \mathcal{A}_j$$

so that

$$c_\tau^k / p_j = \left(\chi^{\rho-1} \left(1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \right) + \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \right)^{\frac{1}{\rho-1}},$$

from which equation (21) follows.

C.2 Log-linearization of the production function

In this paragraph, we derive the log-linear wage regression equation in (17). Starting from the wage equation (15), and imposing $\mathcal{A}_j = \varnothing$ (so that $\Gamma_j = 1$ and $\mathcal{N}_j = \mathcal{T}_j$) for all $j = 1, \ldots, J$, we obtain

$$\log w_j(\boldsymbol{h}) = \log p_j + \sum_{s \in S} \omega_s \log(h_s)$$
$$+ \frac{1}{\rho - 1} \log \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \exp \left(-\eta \sum_{s \in S} \min \left\{ h_s - r_{\tau,s}, 0 \right\}^2 \right)^{\rho-1} \right).$$

Now define the variable $m_\tau \equiv \sum_{s \in S} \min\{h_s - r_{\tau,s}, 0\}^2$ and log-linearize the wage function around $m_\tau = 0$ for all $\tau \in \mathcal{T}_j$. That is, we linearize the wage function around the perfectly matched worker.

A first-order Taylor expansion around this point yields

$$\log \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \exp \left((1-\rho)\, \eta\, m_\tau \right) \right)$$
$$\approx \log \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right) + \eta (1-\rho) \frac{\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} m_\tau}{\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1}}$$
$$= (1-\rho) \left(\eta \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} m_\tau \right)$$

where the second equality follows from $\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} = 1$. Combining the equations above with the definition of m_τ yields equation (17).

D Data

D.1 Task-level Skill Requirements

We elicit a task's skill requirements by replicating O*NET's occupation-level questionnaire on the task-level using OpenAI's GPT-4o. We requested the skill requirement for each of the 19,530 tasks for each of the 35 skill dimensions, resulting in 683,550 independent prompts. As in O*NET, the skill requirements are rated from 1 to 7 and each of the 35 skills have different "level anchors" to indicate the meaning of levels 2, 4, and 6. These anchors, as well as the set of tasks in each occupation, and their descriptions, are taken from the O*NET database. Below, we present the full text of the prompt for the skill *Reading comprehension*, the occupation *Chief Executives*, and the task *"Prepare budgets for approval, including those for funding or implementation of programs."*

> *The occupation [Chief Executives] contains the task: [Prepare budgets for approval, including those for funding or implementation of programs].*
>
> *What level of skill in [reading comprehension] is needed to perform the task in this occupation well?*
>
> *Provide the answers on a scale from 1 to 7, where 2 means [Read step-by-step instructions for completing a form], 4 means [Understand an email from management describing new personnel policies], and 6 means [Read a scientific journal article describing surgical procedures].*
>
> *Output only a single integer, valued between 1 and 7. Do not output anything else.*

D.2 AI and Task Augmentation, Automation, Simplification

We model technologies' impact on workers through three distinct channels: augmentation, automation, and simplification. We leverage O*NET's assessment framework and descriptions of occupations, tasks, and skills to generate new data using OpenAI's large language models. In our baseline scenario, we only consider Generative AI. However, we also consider automation by Autonomous Vehicles, and Smart Robots. For automation assessments, we use GPT-5; for augmentation and simplification channels, we use GPT-4o (completed before GPT-5's release).

D.2.1 Augmentation of Tasks

For augmentation assessment, we ask GPT-4o to estimate time savings when workers get access to these technologies. We assess all 19,530 O*NET tasks and the technologies "generative AI," "smart robots," and "autonomous vehicles," resulting in 58,590 prompts that are evaluated independently. We use OpenAI's GPT-4o with temperature between 0.05 and 0.1.

The prompt structure is consistent across technologies, varying only in the technology description. Here we show the full prompt for **Generative AI**:

> *We are conducting a rigorous assessment of the time a worker can save on specific tasks by using Generative AI.*
>
> *1. Description of technology: Generative AI*
>
> *Generative AI refers to AI techniques that learn a representation of artifacts from data, and use it to generate brand-new, unique artifacts that resemble but don't repeat the original data. These artifacts can serve benign or nefarious purposes. Generative AI can produce totally novel content (including text, images, video, audio, structures), computer code, synthetic data, workflows and models of physical objects. Generative AI also can be used in art, drug discovery or material design.*
>
> *2. Description of the worker and the task:*
>
> *Worker's role: [Occupation] with an average level of expertise.*
>
> *Worker's access to tools: Has all the standard tools available to someone in this position. In addition, this worker now gains access to a Generative AI.*
>
> *Worker's task: [Occupational task]*
>
> *3. Question:*
>
> *Estimate the percentage of time that the worker can save by using the described Generative AI to assist with the task.*
>
> *4. Output Format:*
>
> *Provide your answer as a percentage (numeric value between 0 and 100). Do not output an explanation or any additional information. The answer should be a single number representing the estimate.*

For **Smart Robots**, the technology description changes to:

A smart robot is an AI-powered, often-mobile machine designed to autonomously execute one or more physical tasks. These tasks may rely on, or generate, machine learning, which can be incorporated into future activities or support unprecedented conditions. Smart robots can be split into different types based on the tasks/use cases, such as personal, logistics and industrial.

For **Autonomous Vehicles**, the description is:

An autonomous vehicle is one that can drive itself from a starting point to a predetermined destination in "autopilot" mode using various in-vehicle technologies and sensors, including adaptive cruise control, active steering (steer by wire), anti-lock braking systems (brake by wire), GPS navigation technology, lasers and radar.

D.2.2 Automation of Tasks

To measure technologies' potential to automate occupational tasks, we follow Eloundou et al. (2024) in using a five-tier rubric ranging from no automation (T0) to full automation (T4) exposure. We assess all 19,530 O*NET tasks and the technologies "generative AI", "smart robots", and "autonomous vehicles", resulting in 58,590 prompts that are evaluated independently. We use OpenAI's GPT-5 with low to medium reasoning effort and temperature between 0.05 and 0.1.

The automation prompt follows Eloundou et al. (2024)'s format, with technology-specific definitions and examples. The prompt for **Generative AI** is:

T Automation Rubric

1. Determine if the occupation/task pair meets the definition of T0 No-Automation Exposure. If it does, label it as T0 and stop.

2. If the occupation/task pair does not meet the definition of T0 No-Automation Exposure, determine if the occupation/task pair meets one of the other definitions and select the label that applies:

- *T4: Full automation exposure*
- *T3: High automation exposure*
- *T2: Moderate automation exposure*

- *T1: Low automation exposure*

Rubric

Generative AI refers to AI techniques that learn a representation of artifacts from data, and use it to generate brand-new, unique artifacts that resemble but don't repeat the original data. These artifacts can serve benign or nefarious purposes. Generative AI can produce totally novel content (including text, images, video, audio, structures), computer code, synthetic data, workflows and models of physical objects. Generative AI also can be used in art, drug discovery or material design.

Assume you are a worker with an average level of expertise in your role trying to complete the given task. You have access to Generative AI as well as any other existing software or computer hardware tools mentioned in the task. You also have access to any commonly available technical tools accessible via a laptop (e.g. a microphone, speakers, etc.). You do not have access to any other physical tools or materials.

Please label the given task according to the rubric below.

***T0 No-Automation Exposure** A class of tasks for which Generative AI cannot conceivably perform any aspect of the task in any manner.*

***T4 Full Automation Exposure** A class of tasks where, in most contexts in which this task is currently performed by a human, Generative AI can complete all aspects of this task with high quality when prompted by a human. The output does not normally require oversight by a human. Oversight is not normally required for tasks labeled T4 because the consequences for failure or inaccuracy are small for this task, human judgment is not necessary to complete this task, and generative models can consistently perform this task with very high quality.*

***T3 High Automation Exposure** A class of tasks where, in most contexts in which this task is currently performed by a human, Generative AI could complete 90-100% of the components of the task when prompted, but the output requires oversight from a human. Oversight is normally required because the consequences for failure or inaccuracy are significant for this task, human judgment is necessary to complete this task, and/or generative models cannot perform all aspects of this task with high quality consistently. These tasks rely almost exclusively on the processing of digital* *information, but human judgment is needed to ensure that any digital outputs from Generative AI are high enough quality to be acceptable for the particular context.*

T2 Moderate Automation Exposure *A class of tasks where, in most contexts in which this task is performed by a human, Generative AI could complete between 50%-90% of the components of the task at high quality. These tasks normally rely heavily on the processing of digital information, but a significant portion of the task also involves actions that Generative AI cannot perform with high quality. These tasks require at least some human action beyond just double-checking generative model outputs (such as interpretation, judgment, human-to-human communication, or physical actions).*

T1 Low Automation Exposure *A class of tasks where, in most contexts in which this task is performed by a human, Generative AI could complete between 0%-50% of the components of the task at high quality. These tasks normally rely only partially on the processing of digital information, while the majority of the task involves actions that Generative AI cannot perform with high quality. A majority of the actions that need to be taken to complete this task require a human to perform the action.*

Definitions

High quality *means someone receiving or reviewing the output would not be able to tell the difference between whether it came from Generative AI or a human. For tasks that require a lot of interaction during the completion of the task (e.g. meetings, negotiations), high quality means the people you were interacting with either would not know or would not care that they were interacting with Generative AI.*

Digital information *or* ***information that can easily be expressed digitally*** *includes but is not limited to text, audio, images, video, PDFs, books, code, and data.*

Annotation Examples

Occupation: Special Education Teachers, Preschool

Task: Develop individual educational plans (IEPs) designed to promote students' educational, physical, or social development.

Automation score (T0/T1/T2/T3/T4): T1

Explanation: GenAI drafts SMART goals, accommodations, progress-monitoring templates, and meeting summaries well. But "develop" in practice includes assessments, legal compliance under IDEA, multi-party negotiation, and parent/team consensus—high-stakes, non-digital work that goes far beyond checking model output. The human does a majority of the task through judgment and human-to-human interaction → <50% automatable at high quality.

Occupation: Construction and Building Inspectors

Task: Inspect and monitor construction sites to ensure adherence to safety standards, building codes, or specifications.

Automation score (T0/T1/T2/T3/T4): T1

Explanation: T1 is right because authority, on-site judgment, and safety liability are inherently human/embodied. Modern AI (CV on photos/video, drone logs, code lookups) can pre-screen and draft reports, but the core task is physical inspection plus enforcement. The given explanation is dated—codes are digital; the real blockers are embodiment, accountability, and legal sign-off.

Occupation: Food and Tobacco Roasting, Baking, and Drying Machine Operators and Tenders

Task: Dump sugar dust from collectors into melting tanks and add water to reclaim sugar lost during processing.

Automation score (T0/T1/T2/T3/T4): T0

Explanation: The task as phrased is purely physical and you've restricted tools to what's on a laptop. GenAI can't perform any part of this task (not "write SOPs" but do the dumping). If the task were broadened to "optimize reclaim procedure / generate checklists," exposure would rise, but for the literal action it's T0.

Occupation: Interpreters and Translators

Task: Refer to reference materials (dictionaries, lexicons, encyclopedias, computerized terminology banks) as needed to ensure translation accuracy.

Automation score (T0/T1/T2/T3/T4): T4

Explanation: GenAI can autonomously retrieve, disambiguate, and compile authoritative references and recommended terms with citations and *confidence scores across most high-resource language pairs. The subtask is fully digital, consistent, and does not normally require human oversight.*

Occupation: Electronics Engineer

Task: Prepare documentation containing information such as confidential descriptions or specifications of proprietary hardware or software, product development or introduction schedules, product costs, or information about product performance weaknesses.

Automation score (T0/T1/T2/T3/T4): T3

Explanation: Given structured inputs (requirements, BOMs, test data), GenAI can generate nearly all documentation variants (design specs, DFMEA summaries, release notes, red-team risk write-ups), maintain traceability tables, and enforce style guides. Consequences and confidentiality mean human review is required, but the model can complete 90–100% of components in digital form.

Occupation: Advanced Practice Psychiatric Nurse

Task: Write prescriptions for psychotropic medications as allowed by state regulations and collaborative practice agreements.

Automation score (T0/T1/T2/T3/T4): T3

Explanation: If the medication choice is already clinically decided, GenAI can populate eRx fields, check interactions/contraindications, map to formularies, and draft prior-auth notes—essentially all digital components. High-stakes legality and safety mandate human sign-off, but execution is 90–100% automatable with oversight. (If the task were "evaluate and decide what to prescribe," exposure would drop; the annotation's verb is "write.")

Now, apply the above rubric to the example below:

Occupation: [Occupation]

Task: [Occupational task]

Automation score (T0/T1/T2/T3/T4):

Explanation:

Below, we highlight the substantive changes for prompts related to **Smart Robots** (other than pure wording changes that replace "generative AI" with "smart robots"):

T Automation Rubric

[...]

Rubric

A smart robot is an AI-powered, often-mobile machine designed to autonomously execute one or more physical tasks. These tasks may rely on, or generate, machine learning, which can be incorporated into future activities or support unprecedented conditions. Smart robots can be split into different types based on the tasks/use cases, such as personal, logistics and industrial.

[...]

Definitions

[...]

Annotation Examples

Occupation: Special Education Teachers, Preschool

Task: Develop individual educational plans (IEPs) designed to promote students' educational, physical, or social development.

Automation score (T0/T1/T2/T3/T4): T1

Explanation: A robot can assist with data capture (sensor-based observations), simple assessments, and pre-filling forms, but developing IEPs requires pedagogical judgment, legal compliance, and multi-party collaboration—most of which remains human.

Occupation: Construction and Building Inspectors

Task: Inspect and monitor construction sites to ensure adherence to safety standards, building codes, or specifications.

Automation score (T0/T1/T2/T3/T4): T3

Explanation: Robots (drones/UGVs) with CV/LiDAR can navigate, capture, measure, compare against BIM/specs, and draft reports (90–100% of components). Human oversight is needed for code interpretation, contractor communication, and legal sign-off.

Occupation: Food and Tobacco Roasting, Baking, and Drying Machine Operators and Tenders

Task: Dump sugar dust from collectors into melting tanks and add water to reclaim sugar lost during processing.

Automation score (T0/T1/T2/T3/T4): *T4*

Explanation: *Repetitive material handling and dosing in a controlled plant are fully automatable with robotic manipulation, sensing, and safety interlocks.*

Occupation: Interpreters and Translators

Task: Refer to reference materials (dictionaries, lexicons, encyclopedias, computerized terminology banks) as needed to ensure translation accuracy.

Automation score (T0/T1/T2/T3/T4): *T4*

Explanation: *Purely digital retrieval/matching. A robot running GenAI can autonomously consult termbases, disambiguate senses, enforce glossaries, and return citations without routine human oversight.*

Occupation: Electronics Engineer

Task: Prepare documentation containing information such as confidential descriptions or specifications of proprietary hardware or software, product development or introduction schedules, product costs, or information about product performance weaknesses.

Automation score (T0/T1/T2/T3/T4): *T3*

Explanation: *With structured inputs (requirements, BOMs, test data), a robot+GenAI stack can draft nearly all documents and maintain traceability. Human review remains for accuracy, confidentiality, and compliance.*

Occupation: Advanced Practice Psychiatric Nurse

Task: Write prescriptions for psychotropic medications as allowed by state regulations and collaborative practice agreements.

Automation score (T0/T1/T2/T3/T4): *T3*

Explanation: *A robot can complete the eRx workflow (populate fields, check interactions, format to payer formularies, draft prior auth), but human authorization/clinical judgment is required; oversight is routine.*

Occupation: Warehouse Workers

Task: Move inventory from receiving dock to storage locations using hand trucks or pallet jacks.

Automation score (T0/T1/T2/T3/T4): *T4*

Explanation: *AMRs/AGVs integrated with WMS can autonomously transport pallets/totes end-to-end in structured warehouses.*

Occupation: Assembly Line Workers

Task: Attach components to products moving along assembly line according to specifications.

Automation score (T0/T1/T2/T3/T4): T4

Explanation: In the typical modern assembly context, robots can complete all attachment steps with consistently high quality. Errors are caught by automated fail-safes/poka-yoke and do not require routine human oversight; technicians intervene only on rare exceptions or maintenance, which is outside the task scope.

Now, apply the above rubric to the example below:

[...]

Below, we highlight the substantive changes for prompts related to **Autonomous Vehicles** (other than pure wording changes that replace "generative AI" with "autonomous vehicles"):

T Automation Rubric

[...]

Rubric

An autonomous vehicle is one that can drive itself from a starting point to a predetermined destination in "autopilot" mode using various in-vehicle technologies and sensors, including adaptive cruise control, active steering (steer by wire), anti-lock braking systems (brake by wire), GPS navigation technology, lasers and radar.

[...]

Definitions

[...]

Annotation Examples

Occupation: Special Education Teachers, Preschool

Task: Develop individual educational plans (IEPs) designed to promote students' educational, physical, or social development.

Automation score (T0/T1/T2/T3/T4): T0

Explanation: Purely cognitive/interpersonal; no driving component for an AV to perform.

Occupation: Construction and Building Inspectors

Task: Inspect and monitor construction sites to ensure adherence to safety standards, building codes, or specifications.

Automation score (T0/T1/T2/T3/T4): T1

Explanation: An AV can transport the inspector to/around sites, but the inspection, judgments, and sign-off remain human; AV contributes a minority transport component.

Occupation: Veterinarians

Task: Drive mobile clinic vans to farms so that health problems can be treated or prevented.

Automation score (T0/T1/T2/T3/T4): T2

Explanation: AVs can perform most road driving to rural sites, but last-meters access (gates, unmarked farm roads, ad-hoc parking/turnarounds) and dynamic on-site constraints often require human intervention. Overall, the AV covers a large portion of the task, but not reliably ≥90% across most contexts.

Occupation: Correctional Officers and Jailers

Task: Drive passenger vehicles and trucks used to transport inmates to other institutions, courtrooms, hospitals, and work sites.

Automation score (T0/T1/T2/T3/T4): T3

Explanation: On-road transport between facilities is highly automatable; AVs can execute routing and vehicle control end-to-end. However, the context is high-stakes (security protocols, perimeter handoffs, incident response), so human oversight remains standard even if the driving component is largely automated.

Occupation: Taxi Drivers and Chauffeurs

Task: Test vehicle equipment, such as lights, brakes, horns, or windshield wipers, to ensure proper operation.

Automation score (T0/T1/T2/T3/T4): T4

Explanation: AV platforms can autonomously run pre-trip self-checks and diagnostics (actuate systems, read sensors/OBD, verify via cameras), producing pass/fail results without routine human oversight in most contexts.

Now, apply the above rubric to the example below:

[...]

D.2.3 Simplification of Tasks

The simplification channel assesses how technologies change the skill requirements for performing tasks. We ask GPT-4o to evaluate skill levels both without and with technology access, allowing us to measure the change in required skills. The prompt asks for both values simultaneously. For example, with **Generative AI**:

> *The occupation [Occupation] contains the task: [Occupational task].*
>
> *Technology name: Generative AI*
>
> *Technology description: Generative AI refers to AI techniques that learn a representation of artifacts from data, and use it to generate brand-new, unique artifacts that resemble but don't repeat the original data. These artifacts can serve benign or nefarious purposes. Generative AI can produce totally novel content (including text, images, video, audio, structures), computer code, synthetic data, workflows and models of physical objects. Generative AI also can be used in art, drug discovery or material design.*
>
> *What level of skill in [Skill] is needed to perform the task in this occupation well WITHOUT access to Generative AI? What level of skill in [Skill] is needed to perform the task in this occupation well WITH access to Generative AI?*
>
> *Provide the answers on a scale from 1 to 7, where 2 means [Skill Level 2 Anchor], 4 means [Skill Level 4 Anchor], and 6 means [Skill Level 6 Anchor].*
>
> *Output only two integers separated by a comma, valued between 1 and 7. The first integer is the skill level WITHOUT access to Generative AI, the second integer is the skill level WITH access to Generative AI. Do not output anything else.*

All 35 O*NET skills and 19,530 O*NET tasks are evaluated independently. Skill level anchors and task descriptions are drawn from O*NET as discussed in Appendix D.1.

This approach allows us to measure both the baseline skill requirements $\boldsymbol{r}_\tau$ and the technology-adjusted requirements $\boldsymbol{r}'_\tau$ in a single API call for each skill and task, improving consistency and reducing potential discrepancies from separate queries. The difference between these two values captures the simplification effect of the technology on task skill requirements.

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